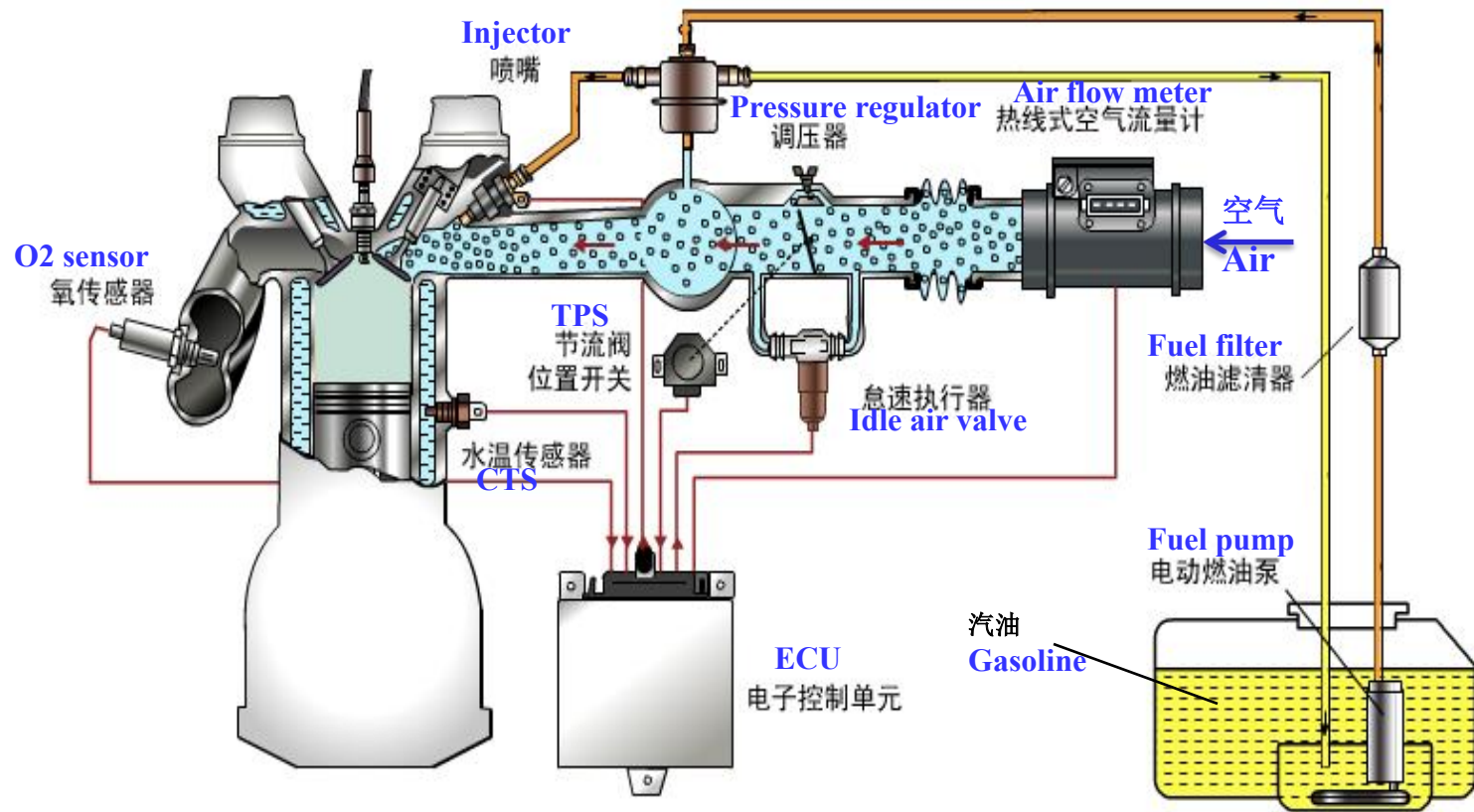


Chapter4 Fuel Supply System For Gasoline Engine

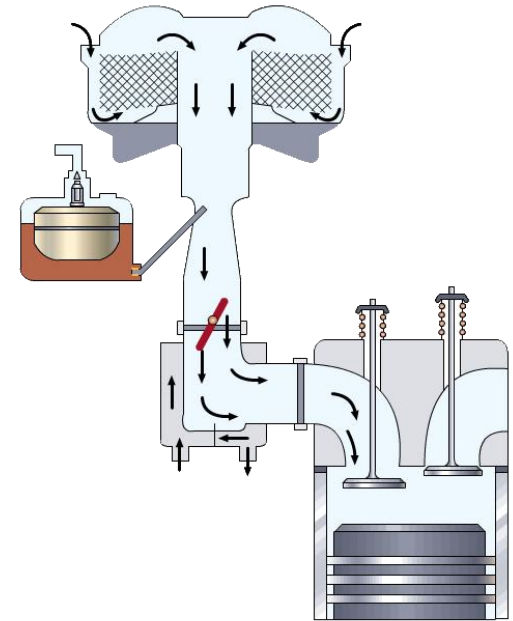
第四章 汽油机燃油供给系统

- The task of fuel supply system is to prepare a certain number of air-fuel mixture of certain concentration according to the working state of engine, then transporting the mixture into cylinder, and finally, pushing the exhaust gas out of cylinder.
- 汽油供给系的任务是根据发动机的不同情况的要求，配制出一定数量和浓度的可燃混合气，供入气缸，最后还要把燃烧后的废气排出气缸。



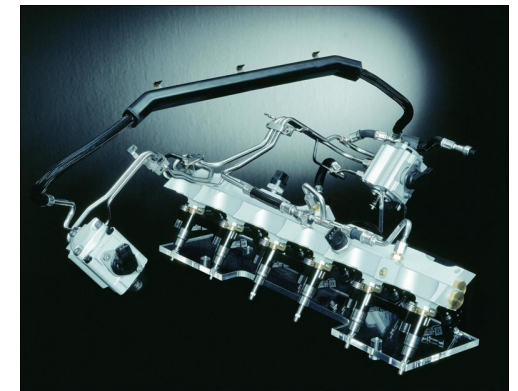
Classification (分类):

Fuel Supply System with Carburetor
传统化油器式燃料供给系



Fuel Supply System with Gasoline Injection by PFI or GDI.

现代汽油喷射式燃料供给系，气道喷射或缸内直喷。



4.1 Fuel of IC Engines (内燃机的燃料)

石油产品
Petroleum
products

汽油 Gasoline

柴油 Diesel

液化石油气 (LPG)

天然气 (CNG)

Essential component of fuel is hydrogen(H) and carbon(C), with a little bit of oxygen(O), nitrogen(N), or sulfur(S)
主要成分是碳、氢，占97~ 98%
含少量硫、氧、氮等。

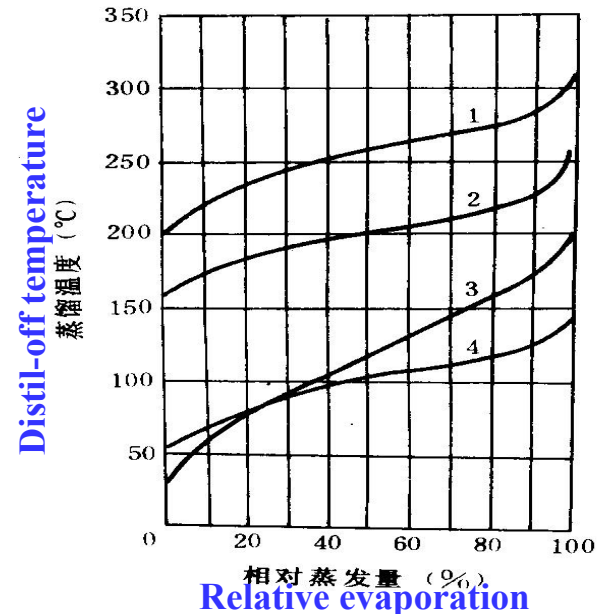
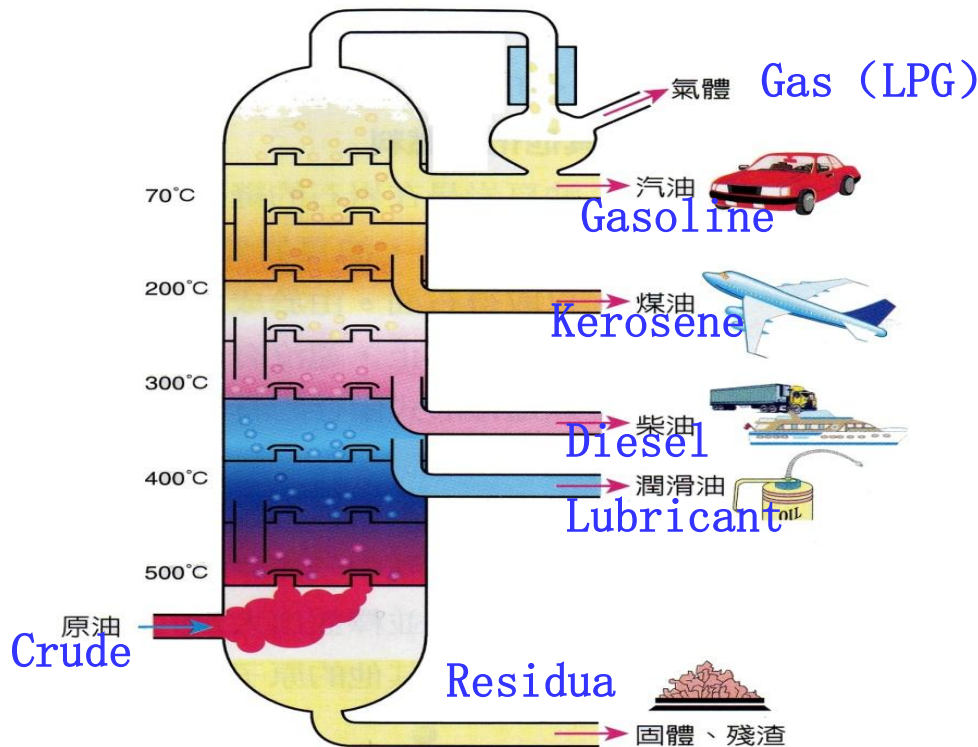


图 3-3 燃料蒸馏曲线的一例

1—轻柴油 2—煤油 3—车用汽油 4—航空汽油
1. Light diesel 2. Kerosene 3. Motor petrol 4. Aviation petrol

- 仅由碳、氢两种元素组成的化合物统称碳氢化合物，简称烃。
- The compound composed only of carbon and hydrogen is named as hydrocarbon.



根据烃分子中碳原子数的不同可构成不同分子量，不同沸点的物质
 Different number of carbon in hydrocarbon molecule means
 different boiling point

C原子数 Number of carbon	沸点 Boiling point(°C)	品种 Species	分子量	性质变化趋势 Changing of nature
C1	常温Room temperature	天然气 Natural gas	16	密度小、易挥发、 稳定性好、易点燃 Smaller density /easy to volatilize/better stability/easy to be ignited 粘度增大、易自燃 Large Viscosity /Easy to spontaneous combustion
C2~C4	常温Room temperature	液化石油气 LPG	16~58	
C5~C11	50~200	汽油 Gasoline	95~120	
C11~C19	180~300	煤油 Kerosene	100~180	
C16~C23	250~360	轻重柴油 Diesel	180~200	
C23	360以上	渣油	220以上	

Performance of Gasoline (汽油的性能指标)

汽油是汽油机的燃料。汽油使用性能的好坏对发动机的动力性、经济性、可靠性和使用寿命都有很大的影响。因此，车用汽油需要满足许多要求。

Performance of gasoline has a great impact on power , economy , reliability and service life of engines. Therefore, there is a number of requirements for motor gasoline.

影响汽油机性能的关键性指标主要是辛烷值和馏程等。

The key figures of the performance of gasoline include **octane number and distillation range.**

1、馏程：用燃油馏出某一百分比的温度范围来表示。

Distillation range : range of temperature in which the fuel is distilled off at a certain percentage.

为了评价燃料的挥发性，以10%、50%和90%的馏出温度作为几个有代表性的点。

In order to evaluate the volatility of the fuel, selecting a series of distilled temperature as representative points, namely, 10%,50% and 90%.

10%馏出温度：体现发动机冷起动性能，表明汽油的蒸发性；

50%馏出温度：体现汽油机的加速性和工作稳定性，表明汽油中间馏份的蒸发能力；

90%馏出温度：体现燃料燃烧的完全性（因为不易蒸发的燃料难以形成良好的混合气，易附着在进气道和缸壁上，而且易破坏缸壁的润滑条件

**10% recovered (distilled) temperature
reflect the cold start performance of engine**

**50 % recovered (distilled) temperature :
reflect the acceleration performance and working stability**

**90 % recovered (distilled) temperature :
reflect the ability of completely burned of fuel**

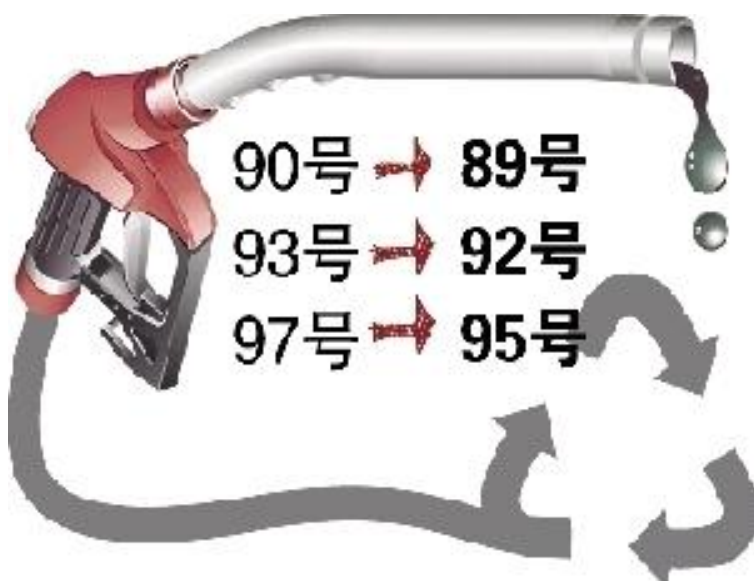
2、热值：指1kg燃料完全燃烧后所产生的热量。

Heat value

means the amount of heat generated by completely burning 1kg fuel

3、**抗爆性**：指汽油在发动机气缸中燃烧时，避免产生爆燃的能力，亦即抗自燃能力。一般用辛烷值表示。辛烷值愈高，抗爆性愈好。

Anti-Knock value: the ability of avoiding to produce deflagration when the gasoline is burning in the cylinder, always expressed by octane number .



汽油的牌号根据汽油的辛烷值确定。压缩比愈高，相应选择的汽油牌号愈高。

The grades of gasoline is determined according to the octane level of gasoline . The higher of the compression ratio is, a higher of gasoline grade should be selected.

4.2 The Characteristic of Air-Fuel Mixture 可燃混合气成分

- 空燃比 α : 可燃混合气中空气质量与燃油质量之比为空燃比
- Air-Fuel ratio α : the ratio of air mass and fuel mass in the mixture
- 过量空气系数 ϕ_a : 燃烧1kg燃油实际供给的空气质量/完全燃烧1kg燃油 的化学计量空气质量
- Excess-air Factor ϕ_a : the ratio of actual air mass and stoichiometric air mass for 1kg fuel.
- $\phi_a < 1$: rich mixture 浓混合气
- $\phi_a = 1$: stoichiometric mixture 理论当量比混合气
- $\phi_a > 1$: lean mixture 稀混合气

Relationship between composition of the mixture and performance of gasoline engine

可燃混合气成分与汽油机性能的关系

可燃混合气成分对发动机性能的影响：

Influences of composition of the mixture to performance of engine:

1) 标准混合气 (stoichiometric mixture) $\phi_a = 1$:

理论上能够完全燃烧的混合气，其中所含的氧气正好使全部燃料燃烧完毕。

2) 稀混合气 (lean mixture) $\phi_a > 1$:

实际上可以完全燃烧的混合气，其中所含的氧气能保证燃料全部燃烧完毕。

3) 浓混合气 (rich mixture) $\phi_a < 1$:

混合气中燃料不能保证完全燃烧，但由于燃料分子密集，火焰传播快，发动机的平均有效压力和功率大。

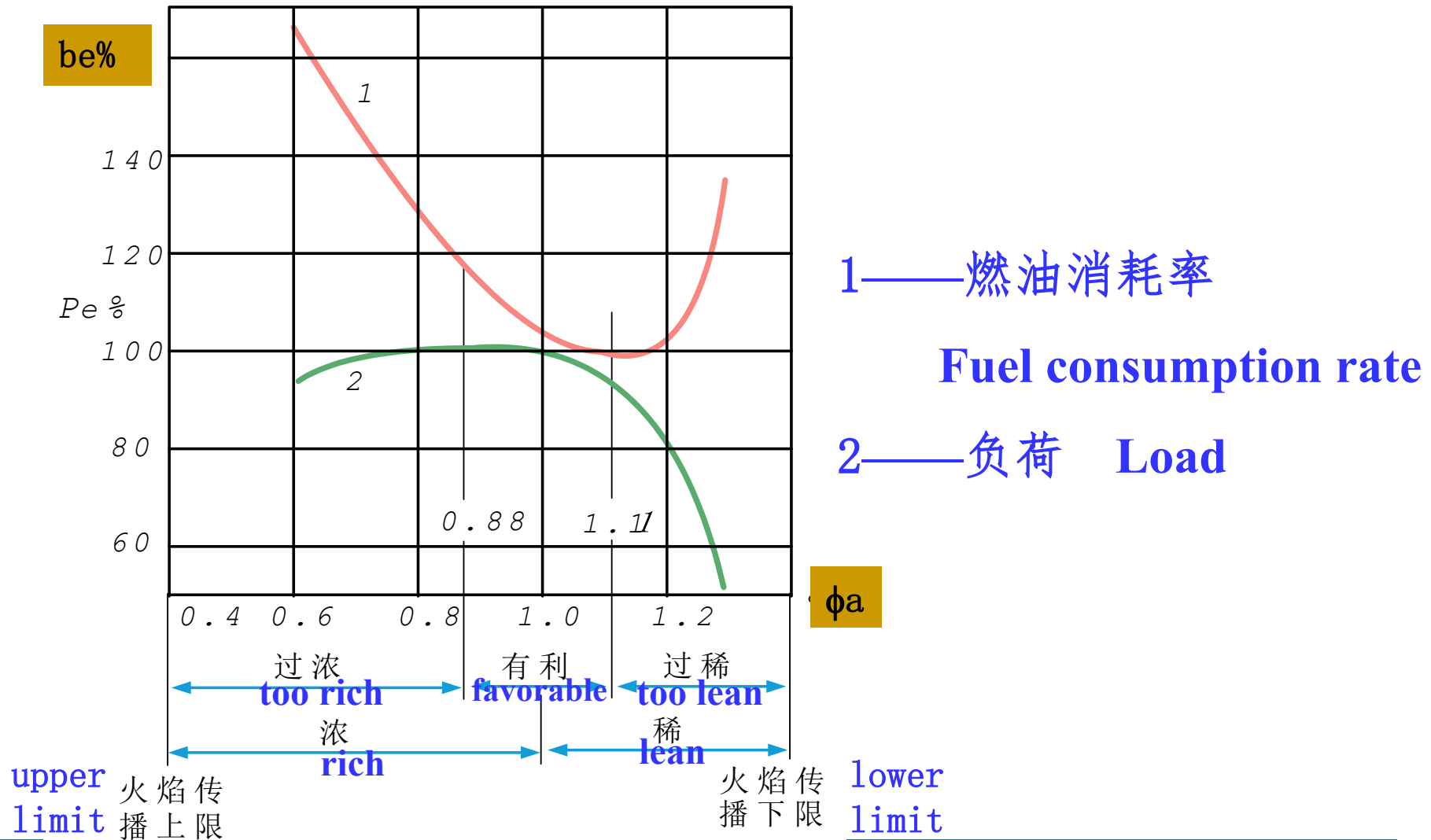
4) 燃烧极限 (flammability limit): $0.4 \leq \phi_a \leq 1.4$

当可燃混合气太稀 ($\phi_a \geq 1.4$) 以及太浓 ($\phi_a \leq 0.4$) 时，虽能点燃，但火焰无法传播，导致发动机运转不稳定，直至熄火。

排放限值和三效催化后处理器的要求，现代汽油机基本上是在标准混合气工作。

Graph of influence of mixture composition on performance

可燃混合气成分对发动机性能的影响曲线图



Influences of mixture concentration on engine performance

混合气的浓度对发动机性能的影响:

混合气种类 Species	ϕ_a	发动机功率 Power	耗油率 b_e	性能 Performance
火焰传播上限 Upper limit	0.4			混合气不燃烧, 发动机不工作 No combustion
过浓混合气 Too rich mixture	0.43-0.87	↓	↑	燃烧室积炭、排气管冒黑烟, 放炮 Deposit/black smoke/blasting
功率混合气 Power mixture	0.88	最大 Largest	↑ _{10-15%}	输出最大功率 Largest output power
标准混合气 Stoichiometric mixture	1.0	↓ _{2%}	↑ _{4%}	
经济混合气 Economic mixture	1.11	↓ _{8%}	最小 Lowest	
过稀混合气 Too lean mixture	1.13 1.33	↓	↑	回火、发动机过热、加速性变坏 Overheating/bad acceleration
火焰传播下限 Lower limit	1.4			混合气不燃烧, 发动机不工作 NO combustion

Requirement for mixture of different engine conditions

发动机各工况对可燃混合气成分的要求（机械式-化油器）

(1)冷起动(Cold start):

极浓混合气 very rich mixture

(2)怠速和小负荷(Idle/small load):

少而浓的混合气 little rich mixture

(3)中等负荷(Medium load):

随节气门的开大,混合气由浓变稀。

With the throttle opening wide , mixture becomes richer.

(4)大负荷(High load):

功率混合气 Power mixture.

(5)加速(Acceleration):

额外供油 extra fuel supply

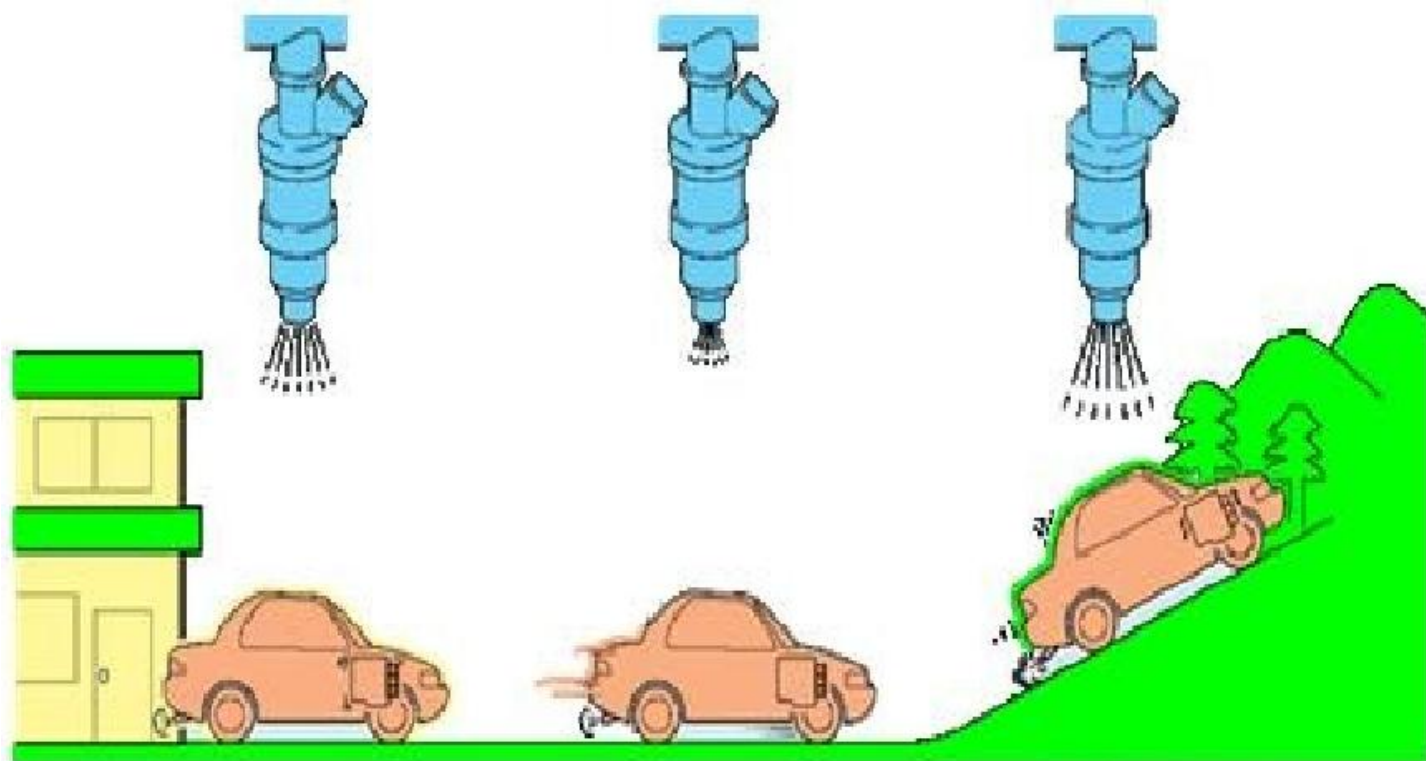
☆怠速: 发动机在对外无功率输出的情况下以最低转速运转,此时混合气燃烧释放的功,只用以克服发动机内部的阻力。

Idle speed : Engine operates at minimum speed without external power output

Requirement for mixture of different engine conditions

发动机各工况对可燃混合气成分的要求（化油器）

工况 Working condition		混合气浓度 Concentration
稳定工况 Stable condition	怠速和小负荷 Idle and low load	$\alpha=0.6\sim0.8$
	中等负荷 Medium	$\alpha=0.9\sim1.1$
	大负荷和全负荷 Big load and full load	$\alpha=0.85\sim0.95$
过渡工况 Transient condition	冷起动 Cold start	$\alpha=0.2\sim0.6$
	暖机 Warmed up	α increases with speed
	加速 Acceleration	Enrich in time



起步和暖机

warm up

等速行驶

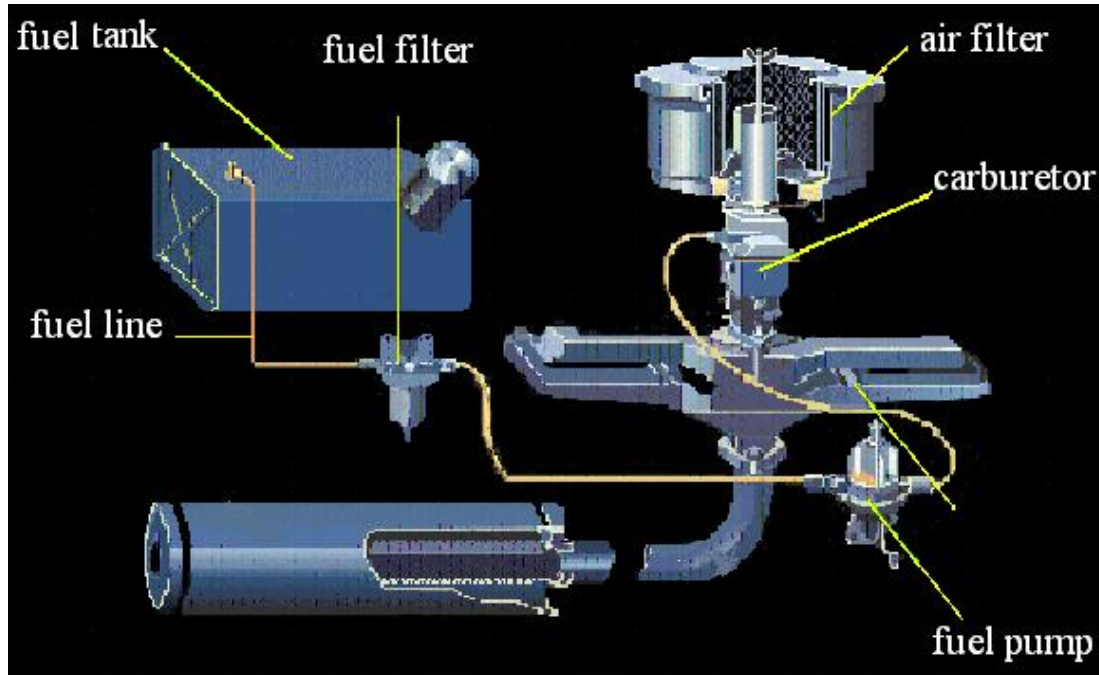
constant speed

高负荷行驶

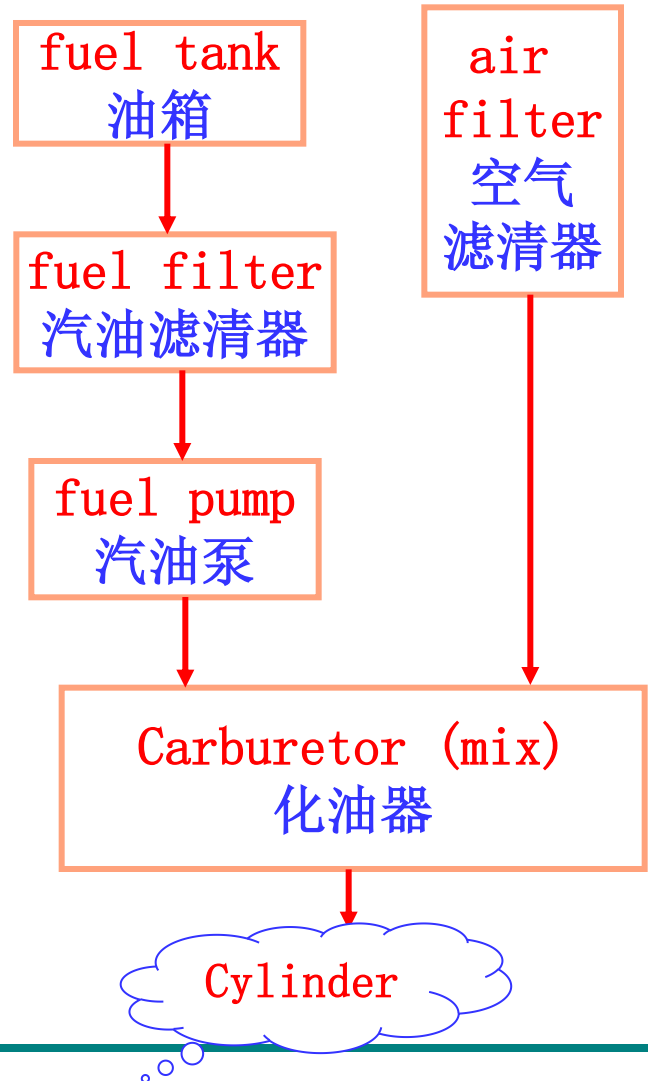
high load

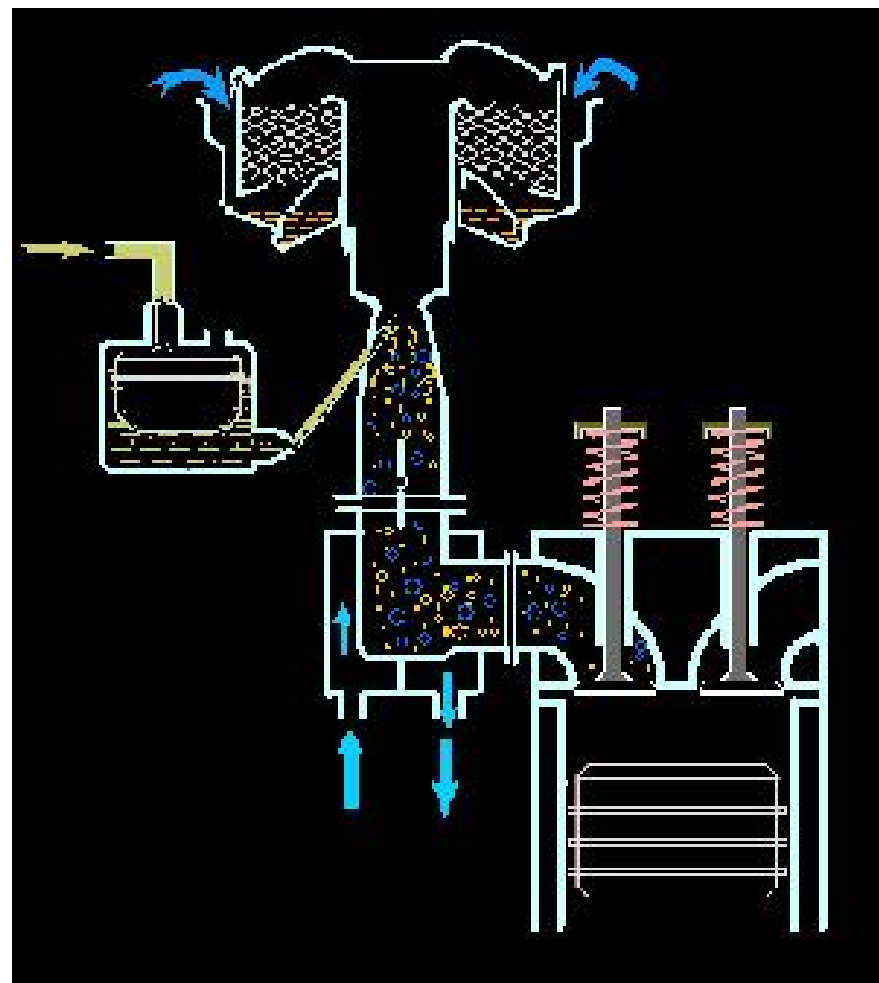
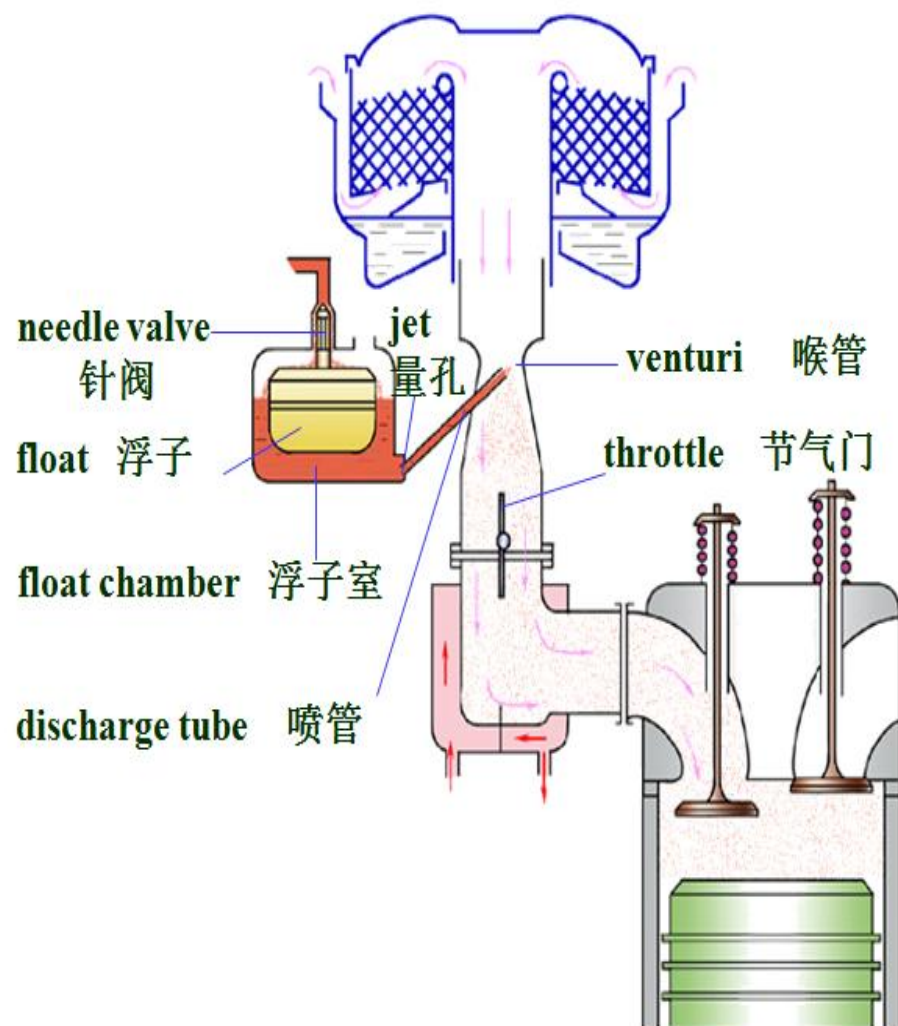


4.3 Carburetor 化油器



No longer preferred in passenger car engines
Used for varieties in certain countries or
two-wheeled vehicle drives.





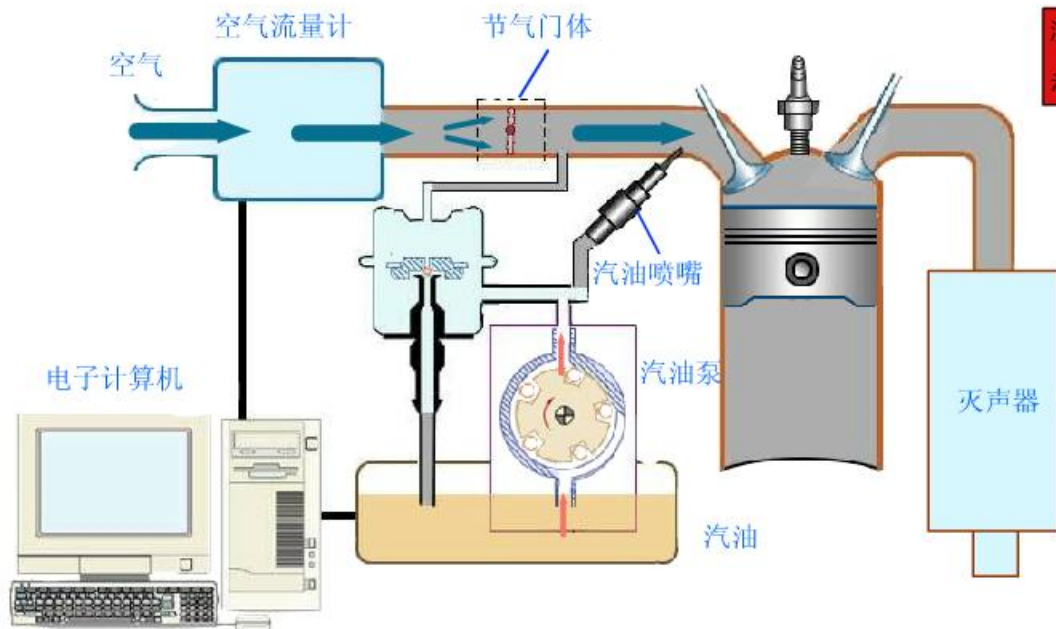
4.4 Fuel Injection System

汽油喷射燃油供给系统

电子控制式汽油喷射系统是通过各种传感器判断发动机的运行状态，发动机ECU对这些信号进行分析、计算、比较、判断后发出喷油脉冲和点火正时指令。

演示

Electronically controlled fuel injection system combines the signals of different sensors to analyze the working state of engine, and then the ECU controls the injector to inject certain amount of fuel and gives suitable ignition timing signal .



History of electronic fuel injection system

电控汽油喷射系统的发展史

1967 年

Bosch公司推出 D 型 **D-Jetronic** Jetronic 模拟式汽油喷射系统。

1973 年

L-Jetronic Bosch公司推出 L 型 Jetronic 的汽油喷射系统，由于采用了测量空气流量的方法控制喷油量，提高了控制精度。同时还开发出机械式汽油喷射系统。

1979 年

Bosch公司推出了集点火与喷油于一体的 Motronic 数字式发动机综合电子控制系统。
在这期间美国 GM公司的 DEFI 、 FORD公司的 EEC，丰田公司的 TCCS 纷纷出场。这些都是综合控制的电子系统。

Motronic

1995 年

美国在轿车上全部采用了电控汽油喷射系统；欧洲的轿车采用汽油喷射系统的占 90% 以上。

目前汽车工业发达的国家在汽油车上均采用汽油喷射系统，以满足日益严格的排放要求。



Compared with Carburetor

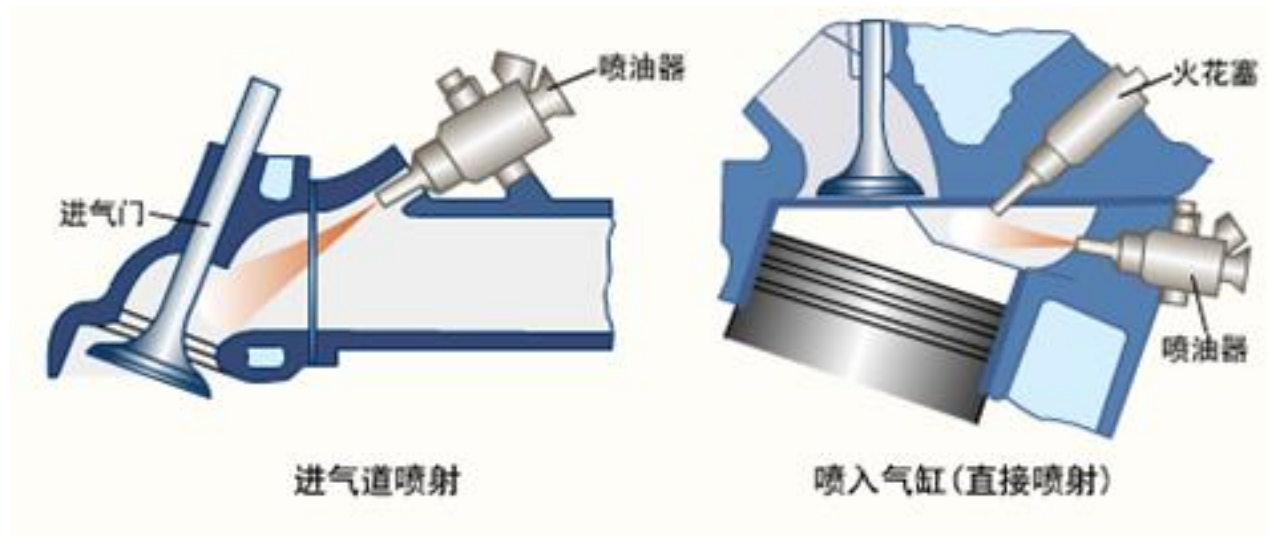
Electronic Fuel Injection	Influence on Automotive			
	Power	Economic	Exhaust	Others
1、在各种工况下均能配制最佳空燃比的混合气 Controlling the air-fuel ratio at ideal value	+	++	++	0
2、取消了喉管，进气阻力减小，提高了发动机的充气效率 No throttle; reducing intake resistance; increasing volumetric efficiency	++	+	+	0
3、燃油雾化质量高；各缸混合气均匀性好 Fuel well atomized; mixture more uniform between cylinders	+	+	++	冷起动性能好 Well cold start
4、空燃比控制系统动态响应快 Fast dynamic response of air-fuel ratio controlling system	+	0	+	加速性能好 Well acceleration performance
5、断油功能 Fuel cut function	0	+	+	0
6、能根据进气温度、海拔高度变化，对空燃比进行修正 Modifying air-fuel ratio according to air temperature and altitude	0	+	+	0

Category of fuel injection system

汽油喷射系统的分类

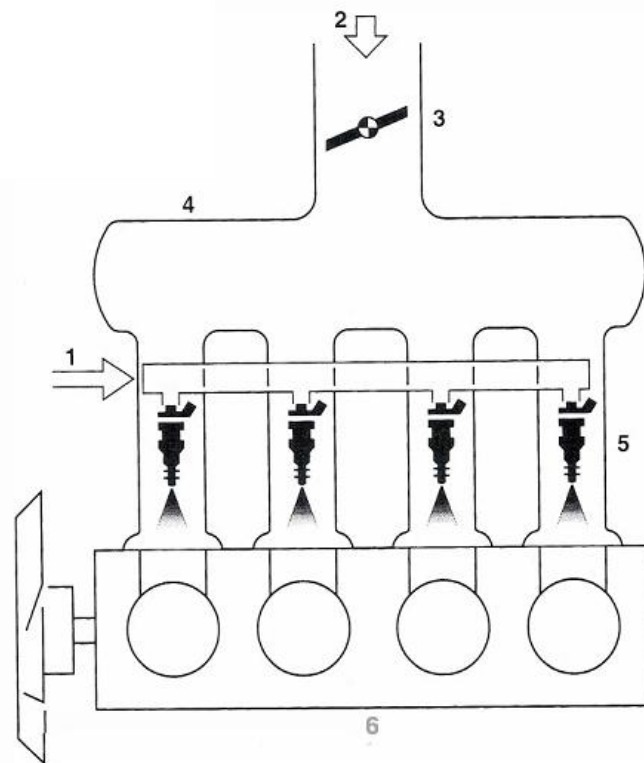
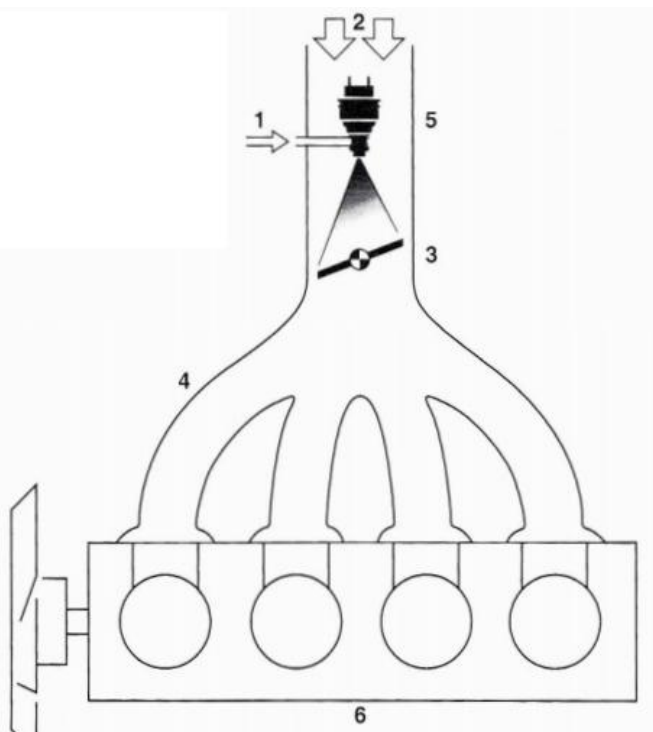
Gasoline injection system can be classified into port fuel injection (PFI) and gasoline direct injection (GDI). The former needs much lower injection pressure than the latter.

汽油喷射按喷射位置分为进气道喷射和缸内喷射两种。前者是低压喷射，后者前者是高压喷射。



1. 按喷射系统执行机构分类:

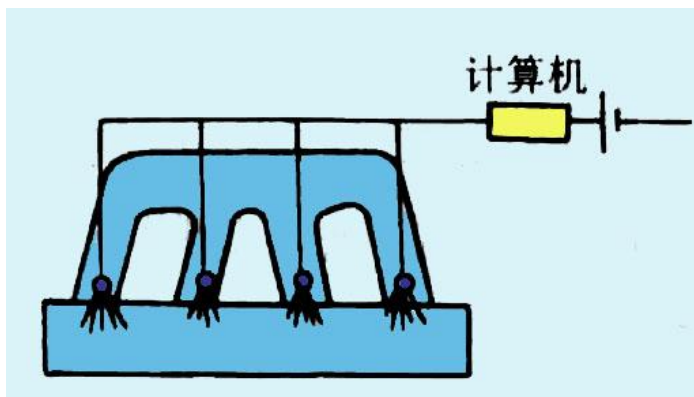
SPI—Single Point Injection (单点喷射) MPI—Multi Point Injection (多点喷射)



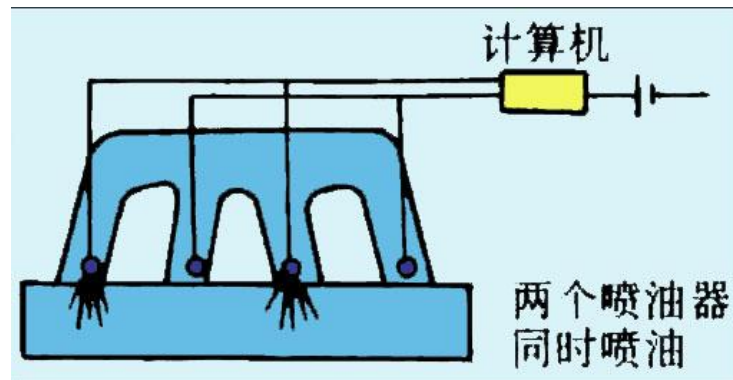
1.Fuel 2.Air 3.Throttle valve 4.Intake manifold 5.Injector 6.Engine

2. 按喷射方式分类: **Classified by injection mode**

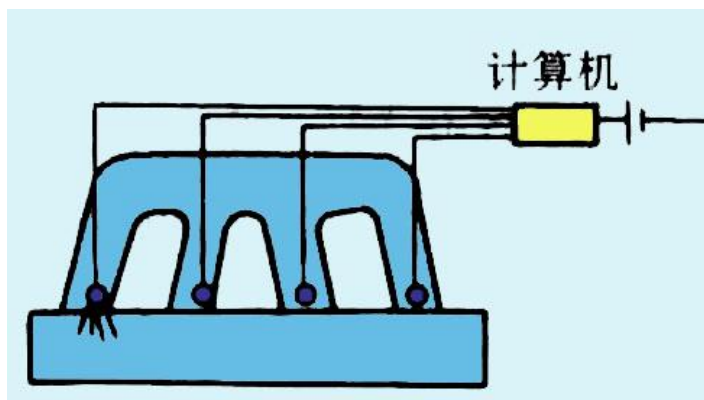
Simultaneous injection 同时喷射



Grouping injection 分组喷射

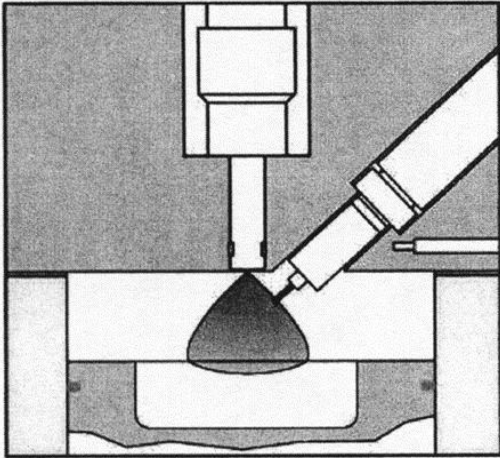


Sequential injection 顺序喷射

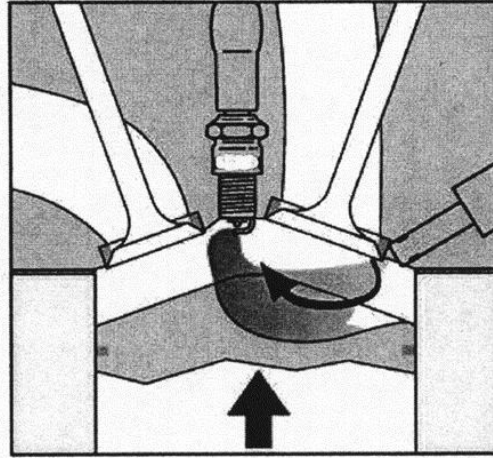


Fuel Injection for GDI

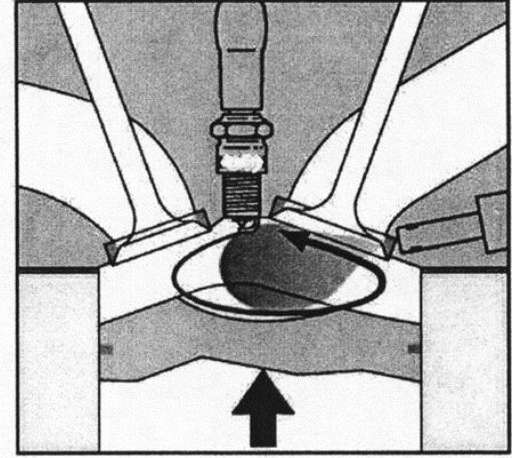
Spray-guided



Wall-guided



Air-guided

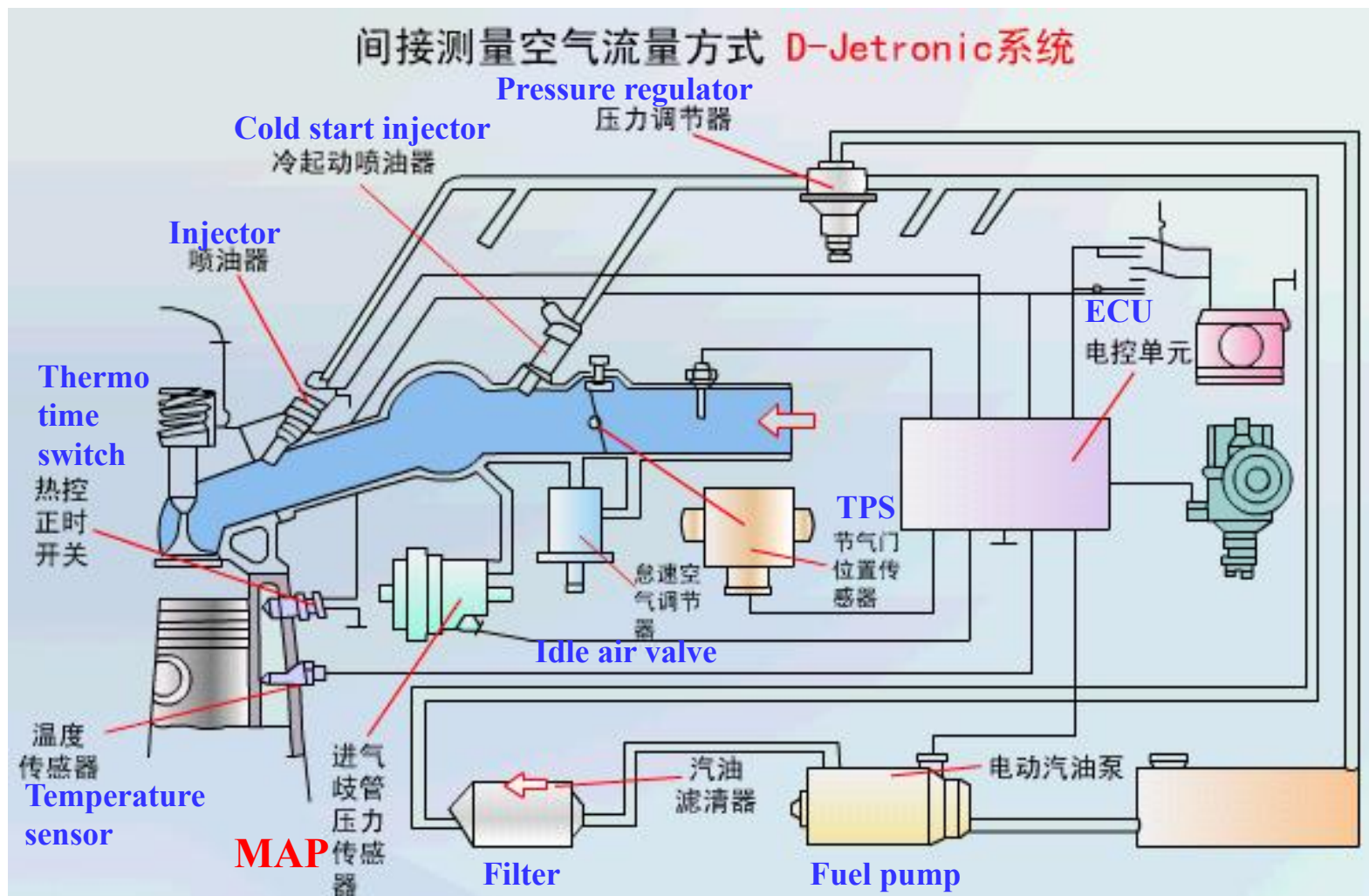


3. 按空气量的计量方式分类：

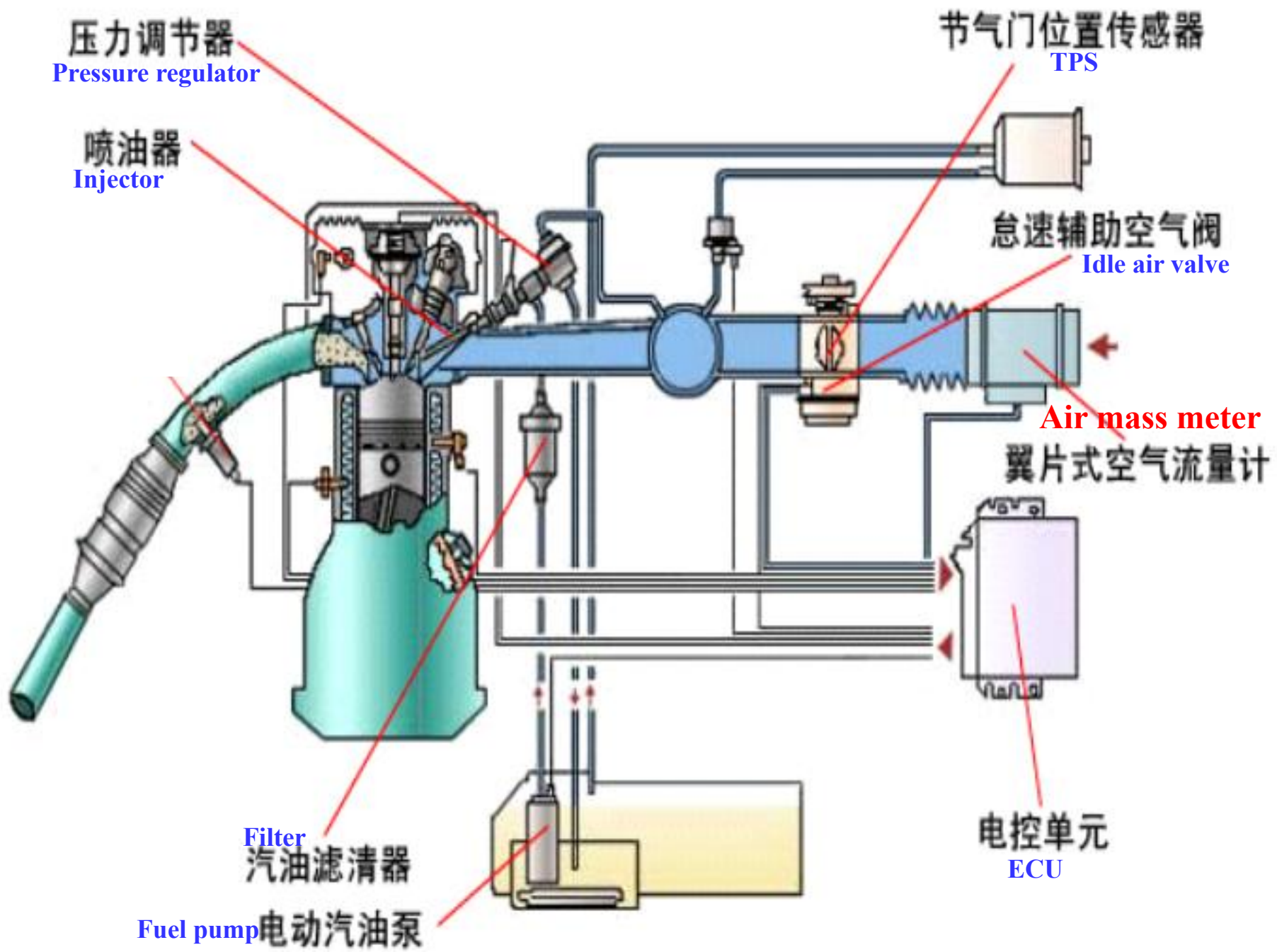
(Classified by the way of measuring air mass)

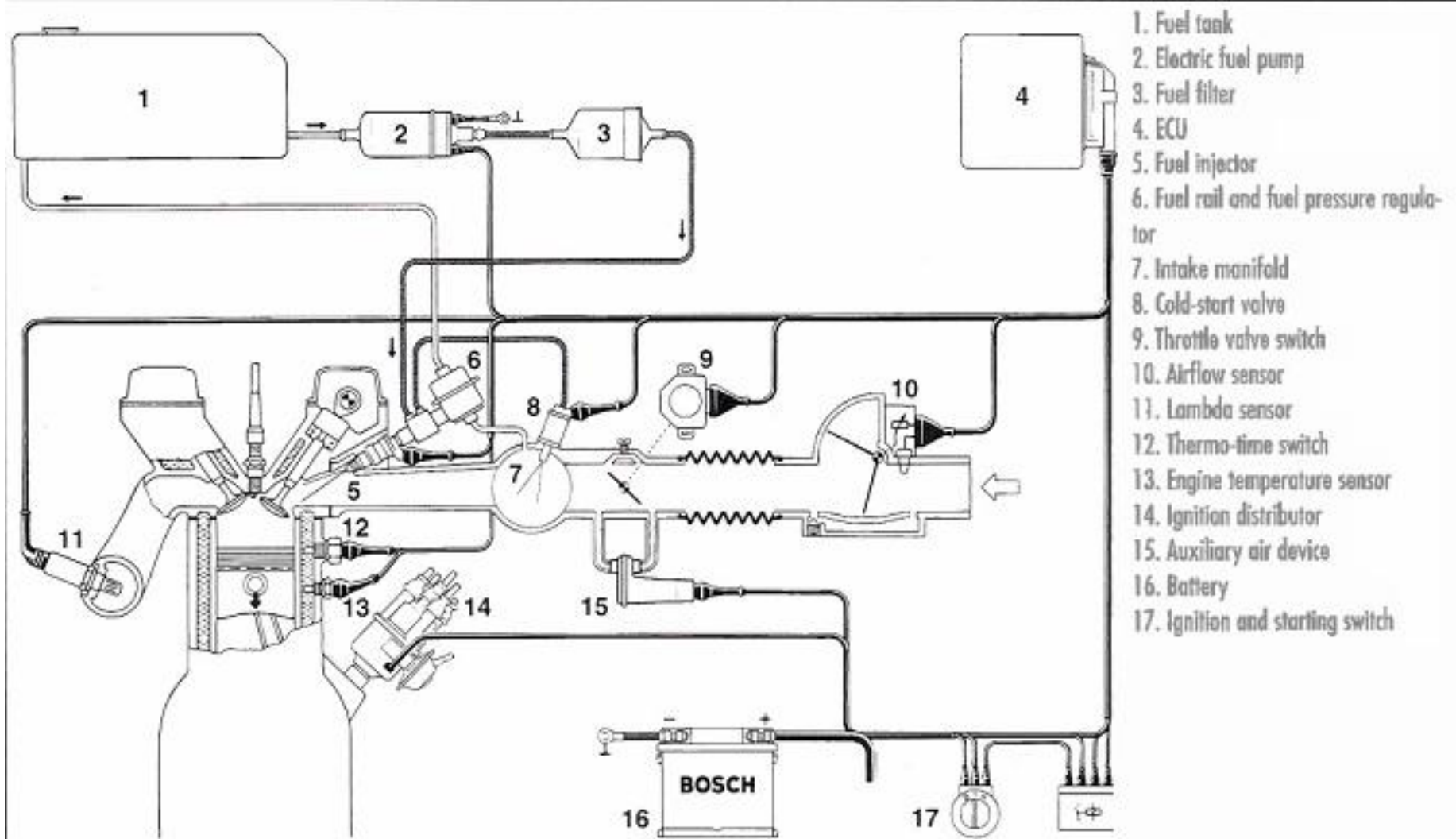
D型电控燃油喷射系统 D-type

L型电控燃油喷射系统 L-type



直接测量空气流量方式 LH-jetronic系统

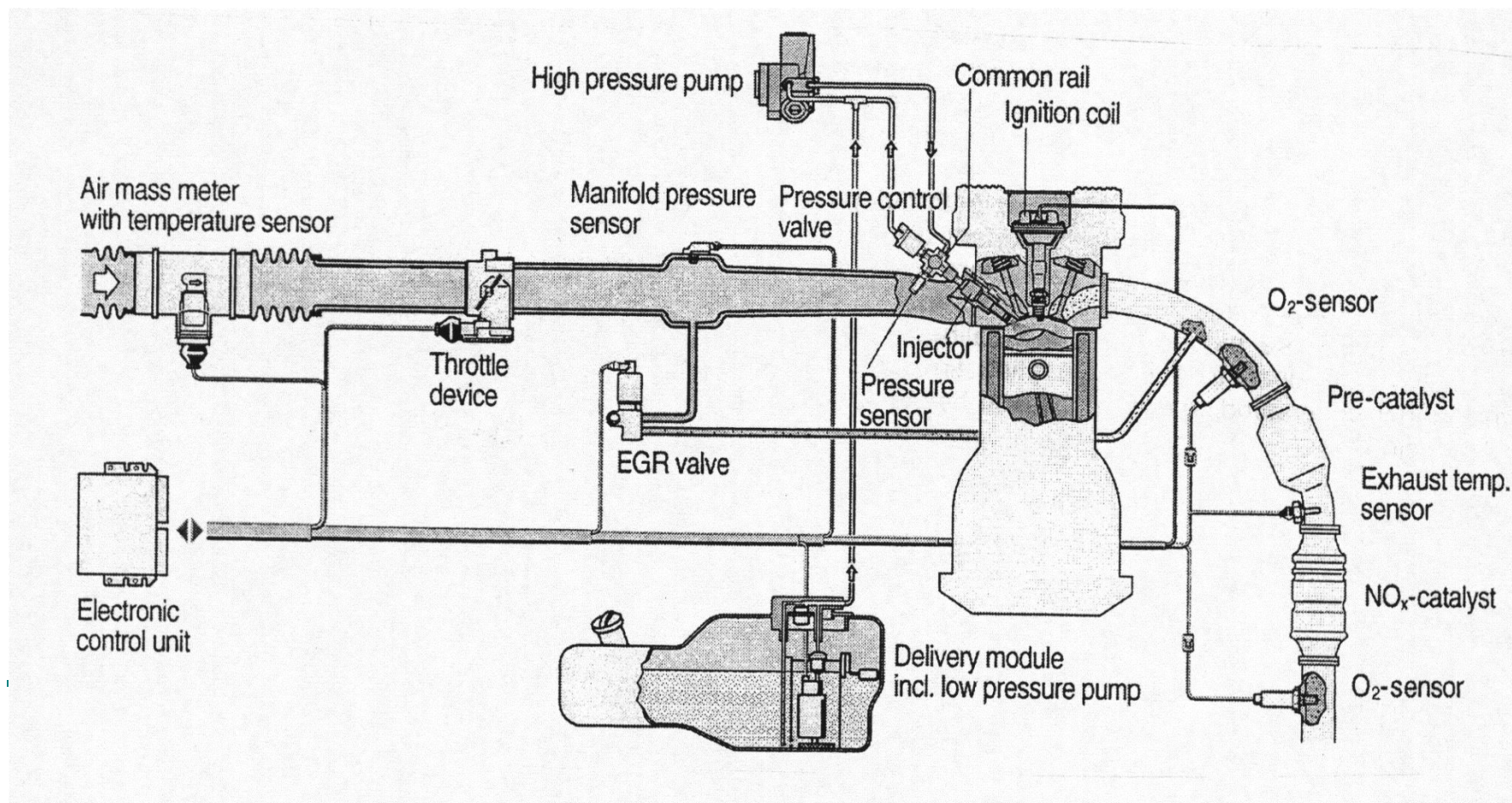




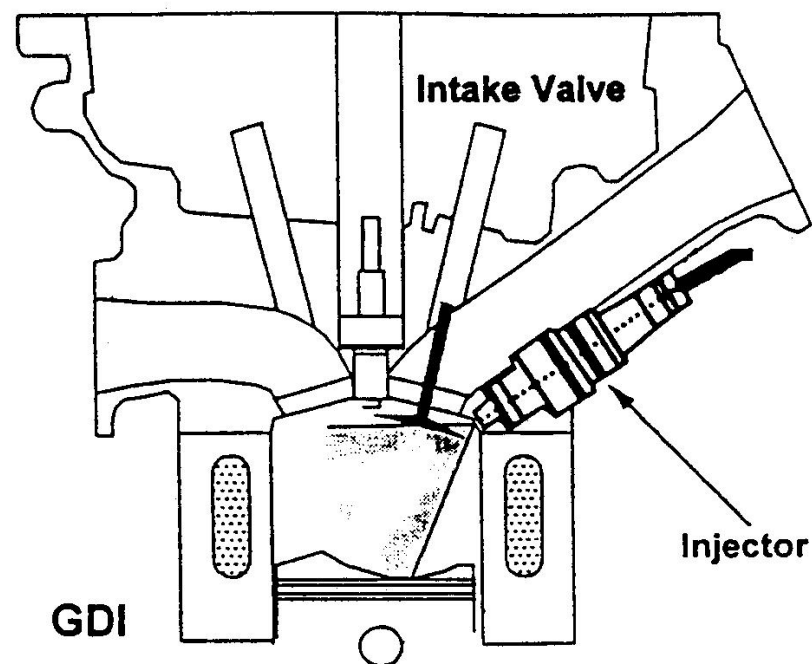
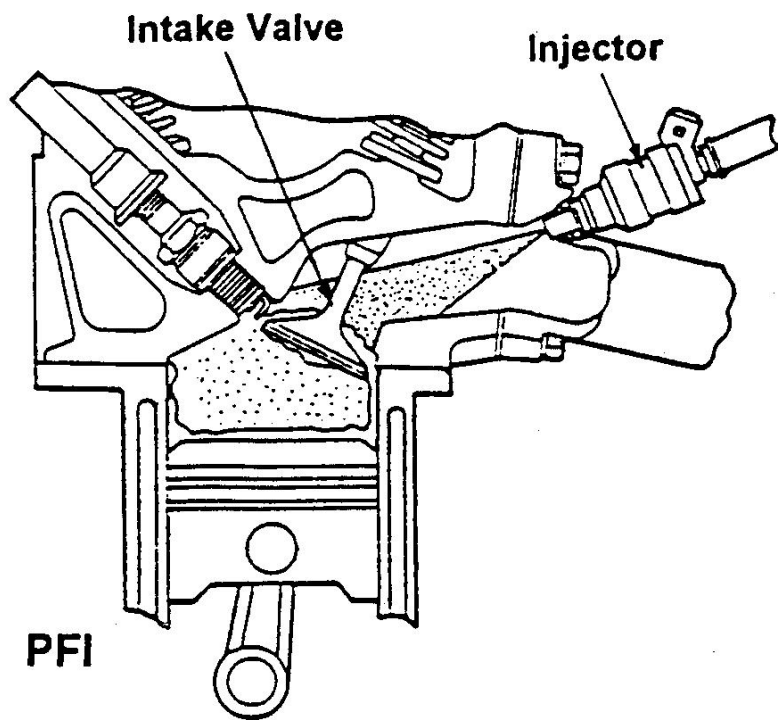
Schematic diagram of an L-Jetronic system with lambda (oxygen sensor) control.

Gasoline Direct Injection versus Port Fuel Injection

Layout of a typical G-DI Engine System



GDI Versus PFI

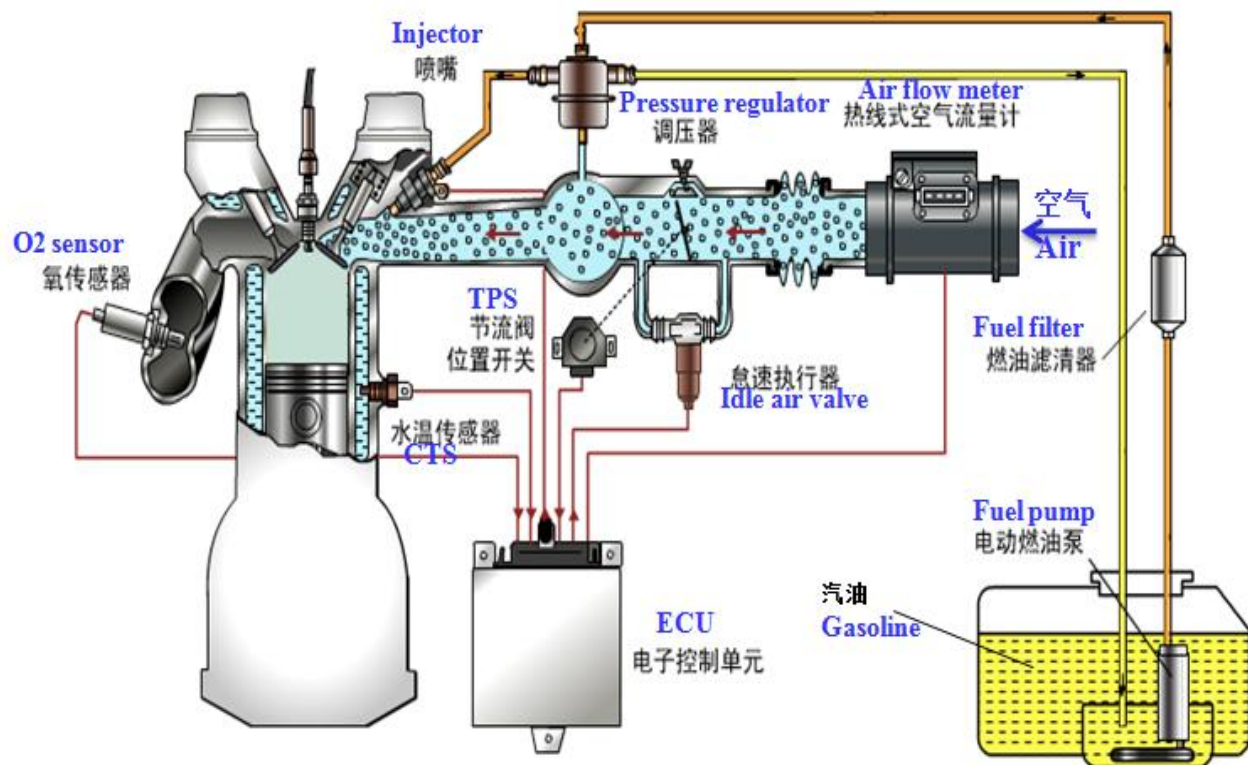


4.5 Component of Electronic Fuel Injection System

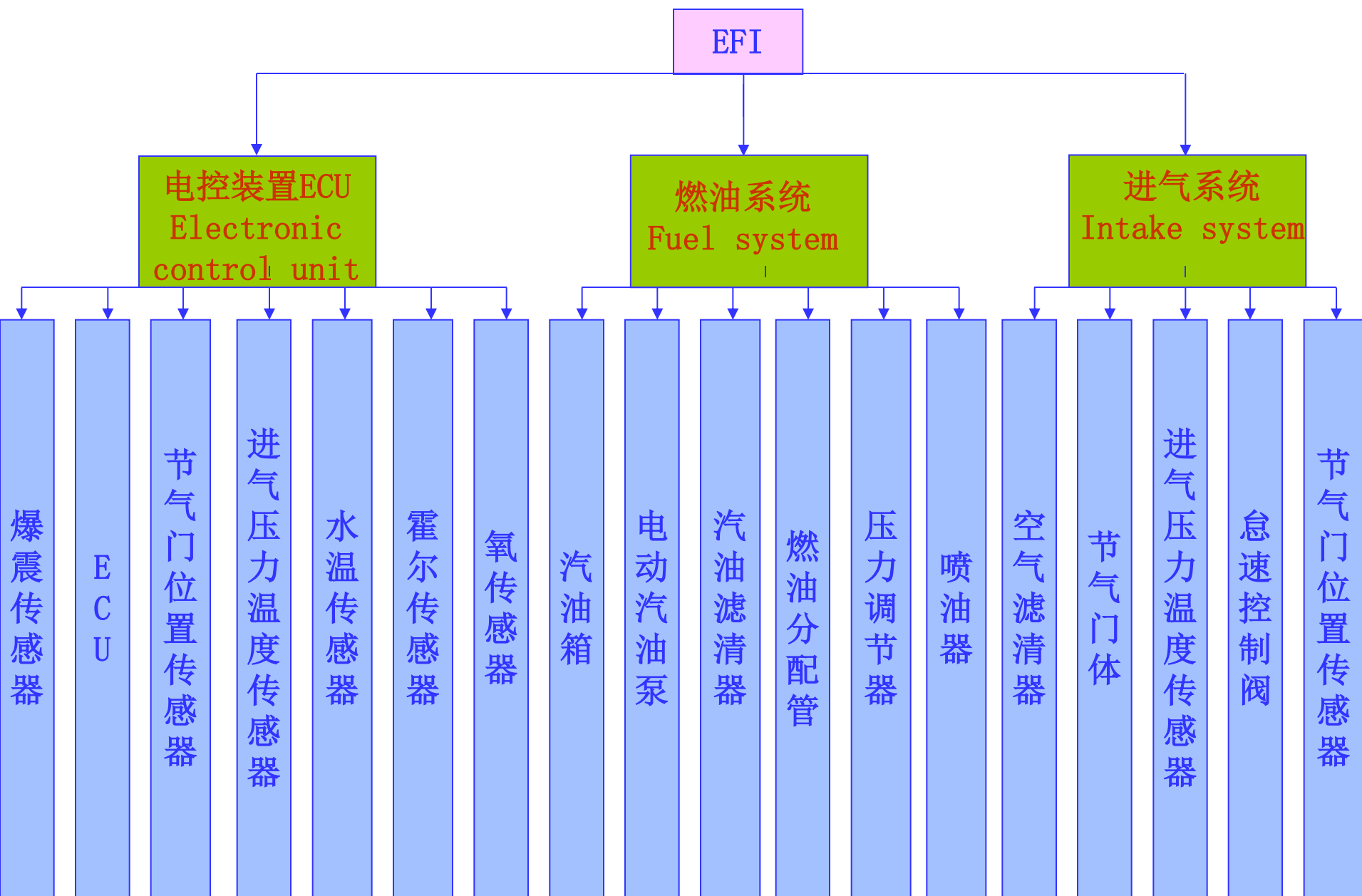
电控燃油喷射(EFI)系统基本组成

电控燃油喷射系统一般由三个子系统组成。

- 1、空气供给系统
- 1. Air-supply system
- 2、燃油供给系统
- 2. Fuel-supply system
- 3、电子控制系统
- 3. Electronically controlling system



Video



Electronic System

ECU

Knock Sensor

Throttle Position Sensor

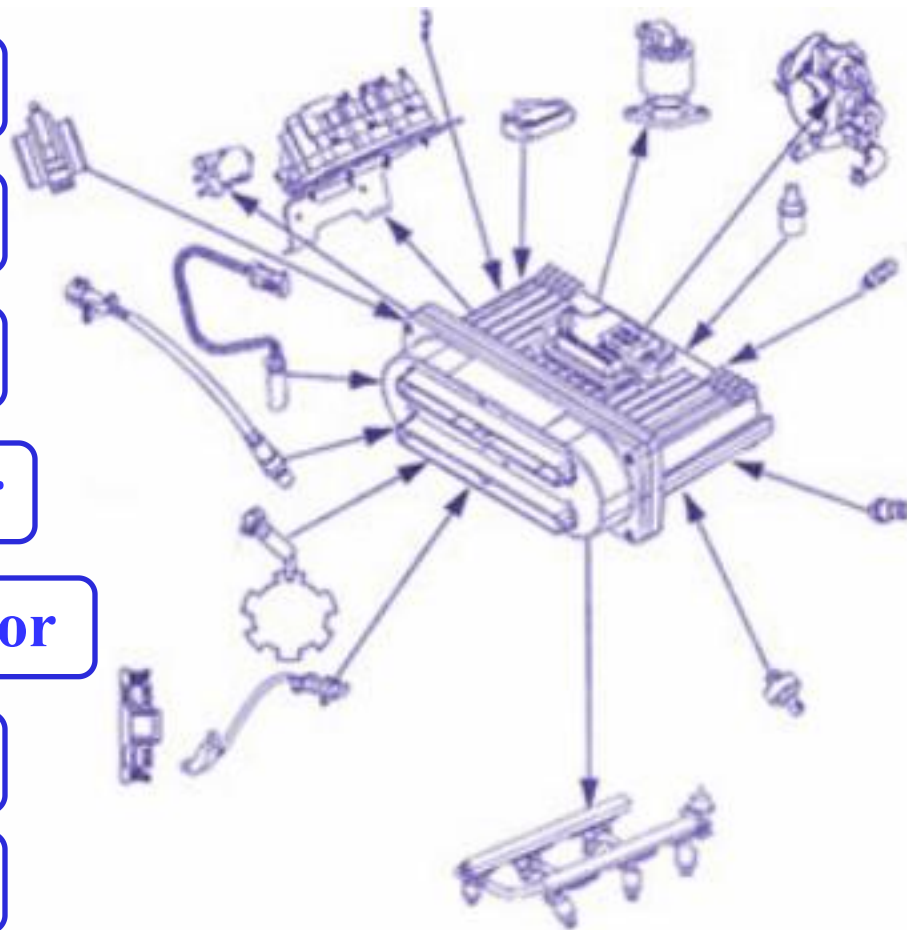
Intake Air Pressure Sensor

Intake Air temperature Sensor

Temperature Sensor

Hall Sensor

Lambda Sensor



Fuel System

Fuel Tank

Electronic Fuel Pump

Fuel Filter

Fuel Rail

Pressure Regulator

Injector

Intake System

Air Filter

Throttle Body

Idle Controlling Valve

Intake Manifold Pressure Sensor

Intake Air Temperature Sensor

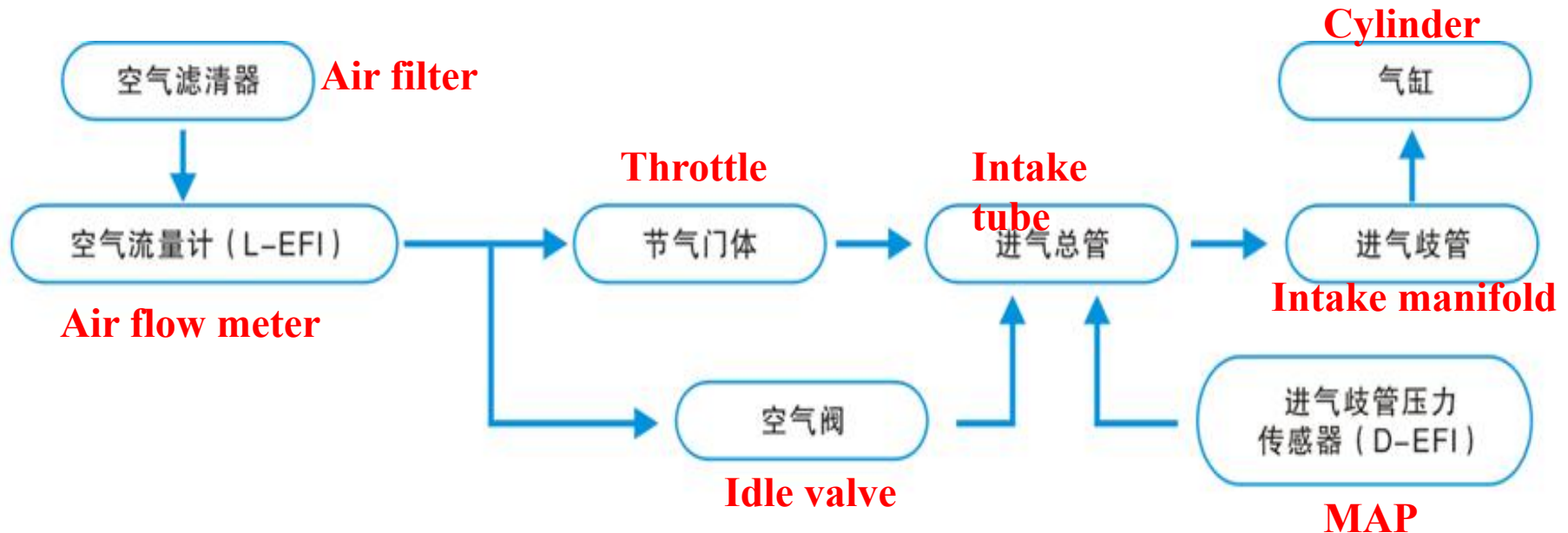
Throttle Position Sensor

Air-supply System

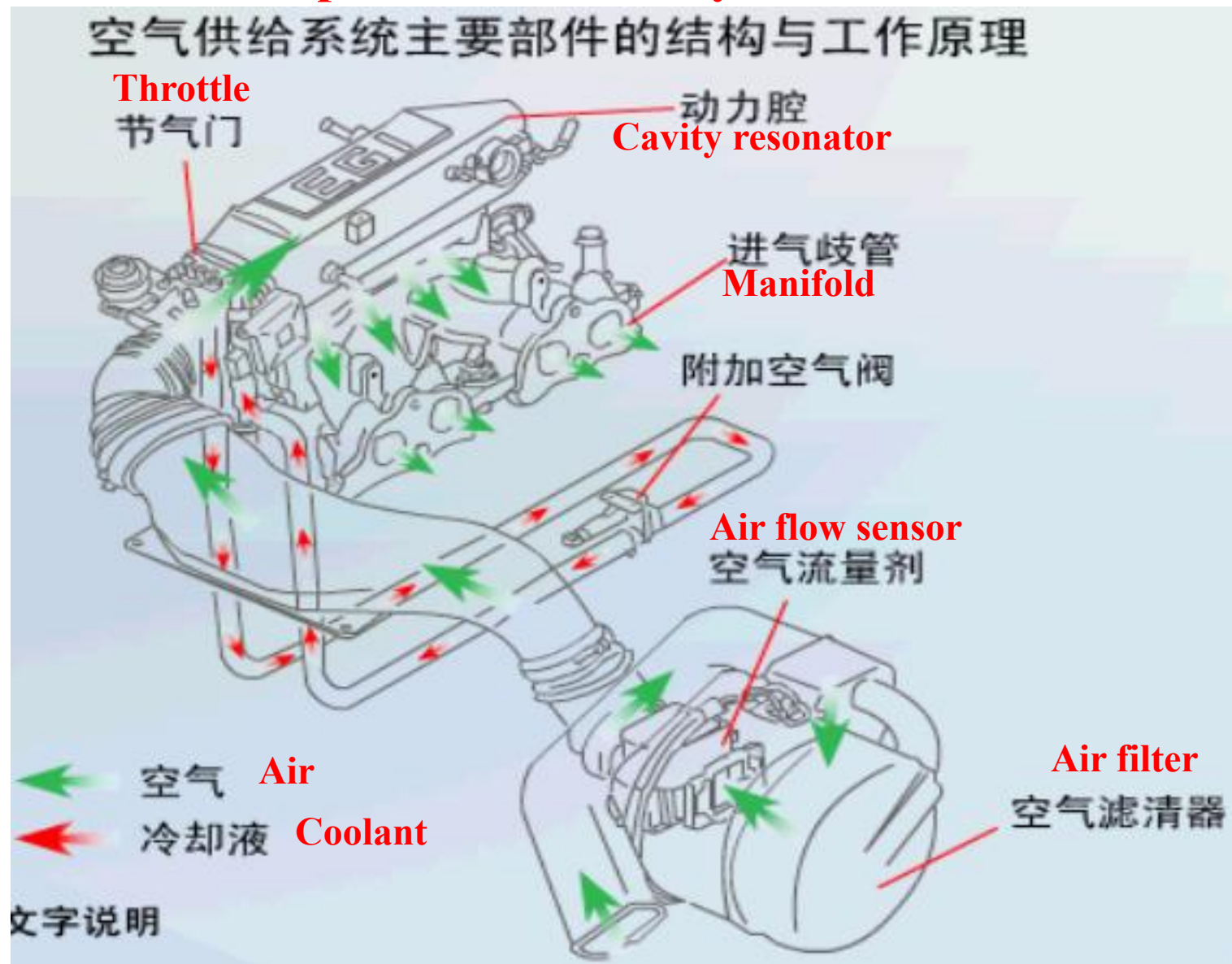
空气供给系统

空气供给系统的作用是向发动机提供新鲜的空气。它主要由空气滤清器、空气流量计、节气门、进气总管、进气歧管和怠速空气控制阀等组成。

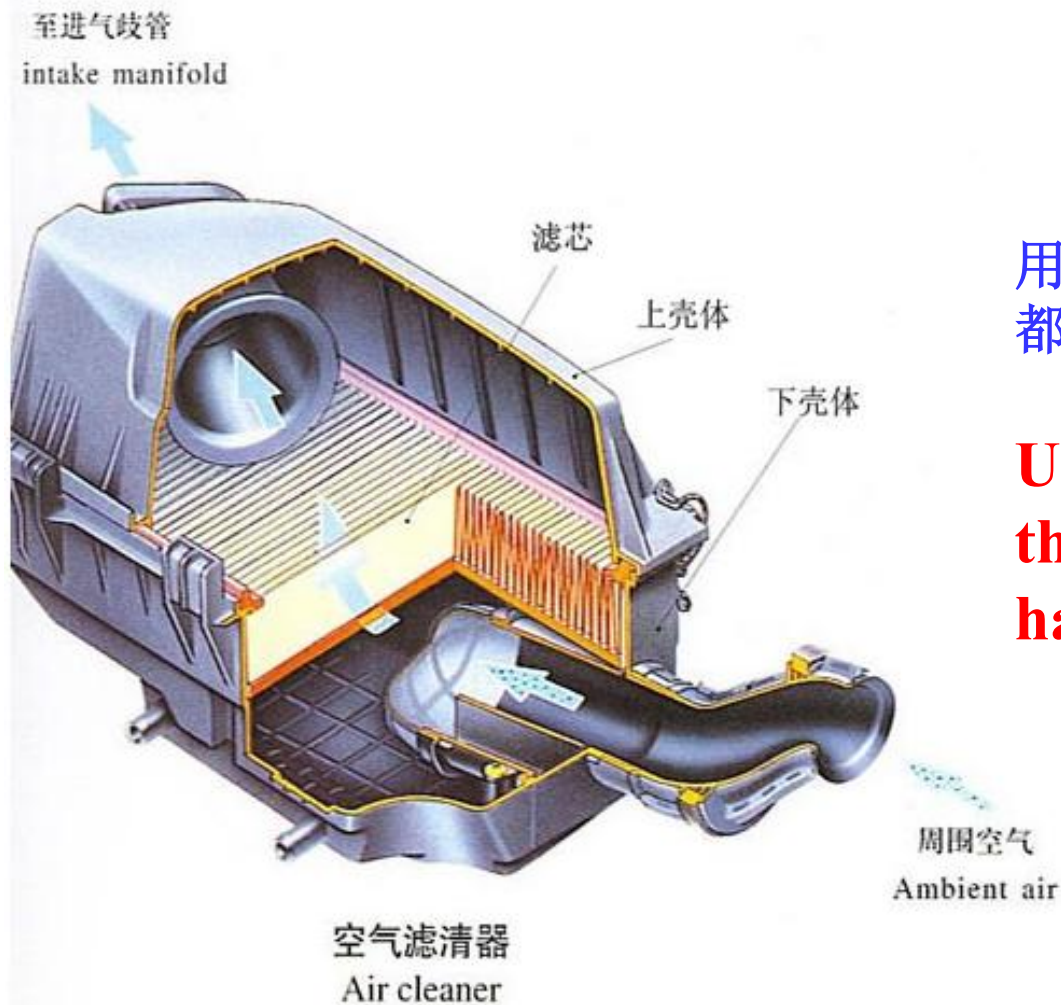
The role of the air-supply system is to provide fresh air to the engine. It mainly consists of the air filter, the air flow meter , throttle , intake tube , intake manifold and the idle air control valve and other components .



Component of intake system



Air Cleaner 空气滤清器

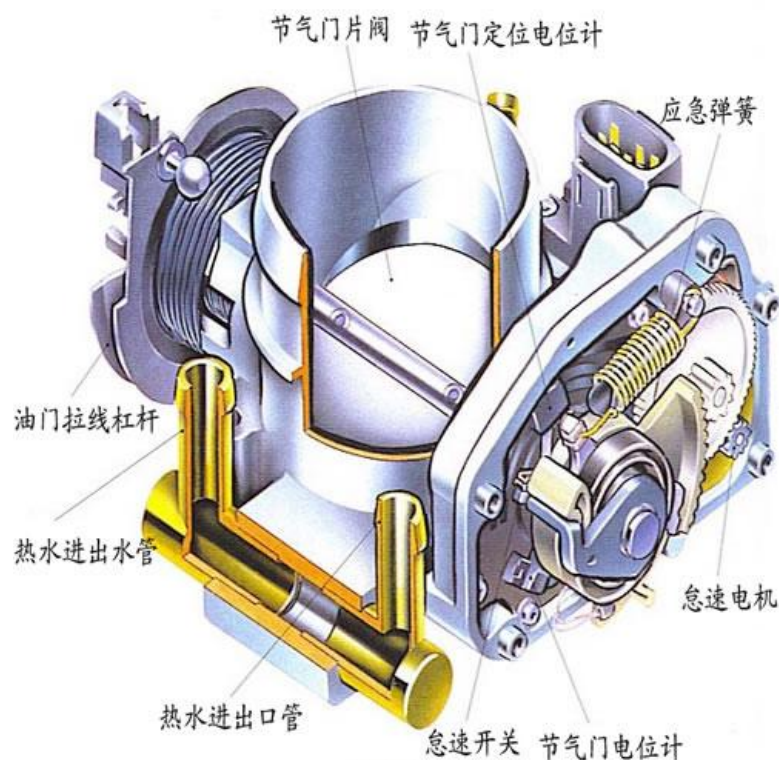


用于滤除空气中的灰尘，一般都为纸质滤芯。

Used to filter the dust in the air, and generally have a paper filter element.

Throttle device 节气门体

- 节气门体安装在进气管中，来控制发动机正常工况下的进气量。
- Throttle body is mounted in the intake manifold, to control the mass of intake air at normal condition.



主要由节气门、位置传感器和怠速空气道等组成

- The throttle body mainly includes throttle, position sensor and idle air valve.

Air Mass Sensor 空气流量计

空气流量计的功用是测量进入发动机的空气流量，并将测量的结果转换为电信号传输给电控单元。

In order to be able to determine the air mass flow drawn in by the engine , either a manifold pressure sensor(MAP) or an air mass sensor is employed. The output signal serves predominantly for controlling the fuel volume as an input parameter for the ignition map and for determining the exhaust-gas recirculation rate.

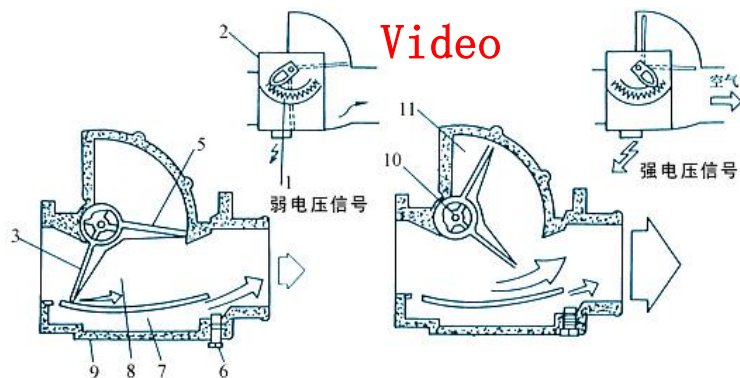
空气流量计有多种形式，如电位计式、热线式、热膜式和涡流式等。

There are a variety of forms , such as potentiometer-type, hot-wire type, hot-film and swirl type .



空气流量计

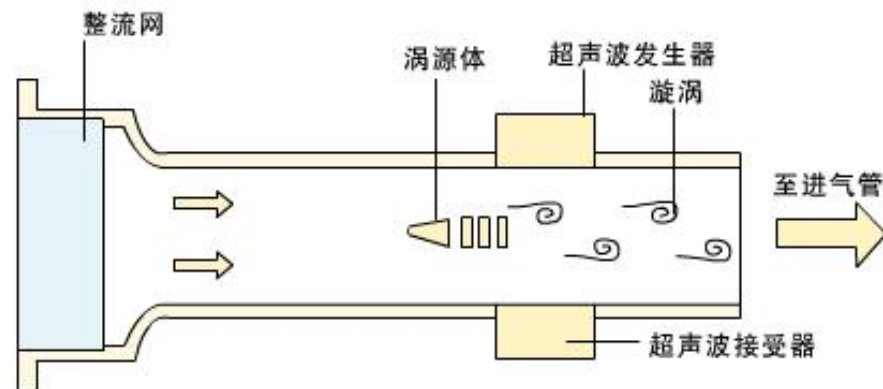
电位计式空气流量计 potentiometer -type



翼片式空气流量计工作原理

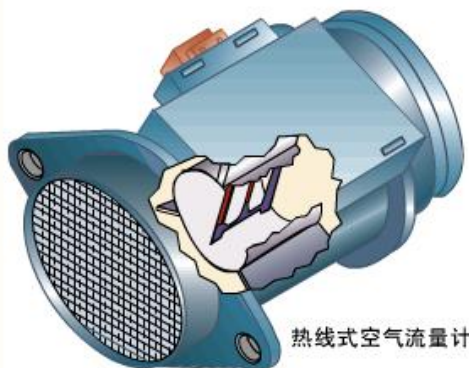
1-卷簧；2-电位计；3-翼片；4-进气温度传感器；5-缓冲片；6-旁通空气调节螺钉；7-旁通空气道；8-主流道；9-空气流量计壳体；10-销轴；11-缓冲室

卡门涡流式空气流量计 Swirl type

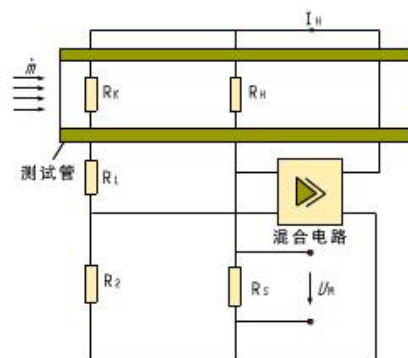


卡门涡流式空气流量计

热线式空气流量计 Hot-wire type

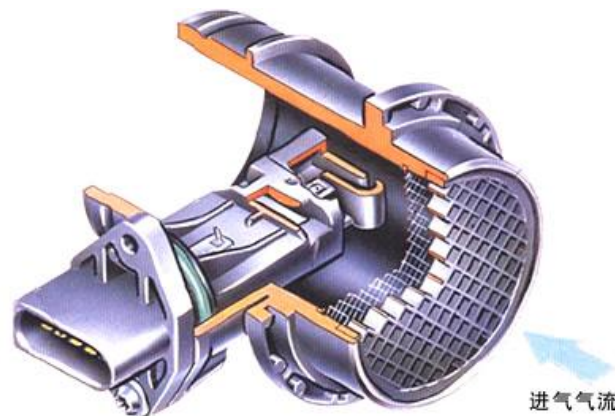


热线式空气流量计



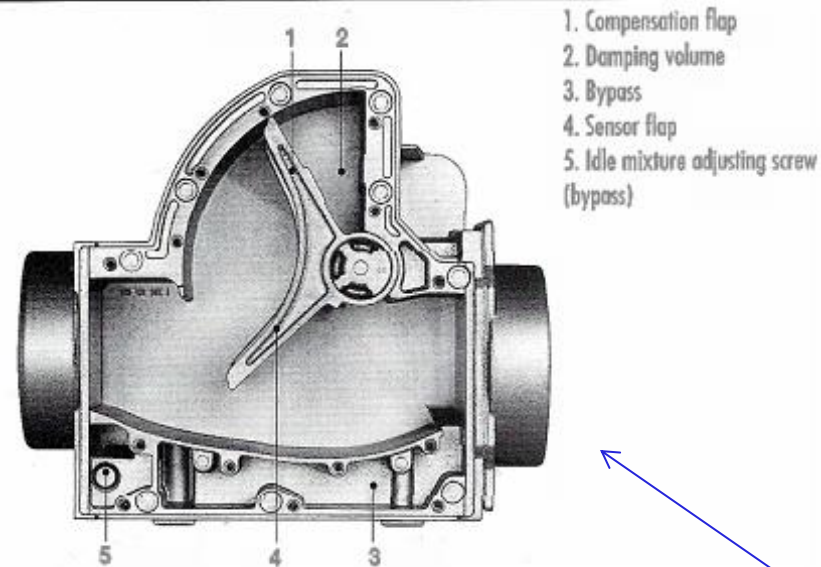
热线式空气流量计电路

热膜式空气流量计 Hot-film type

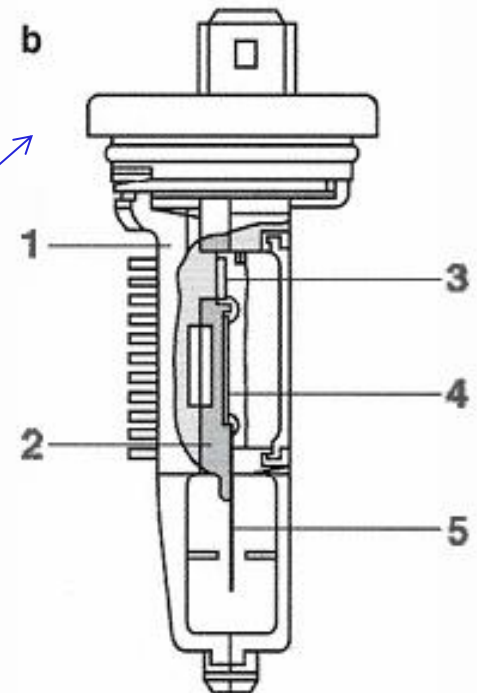
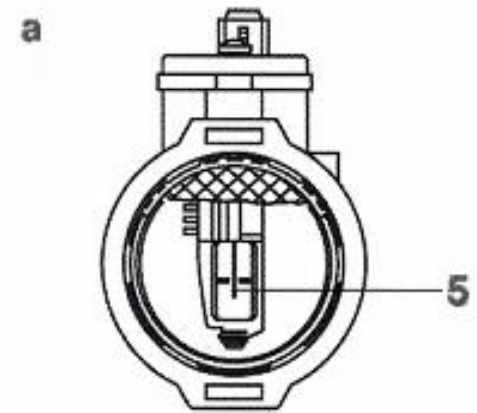


热膜式空气流量计及其桥式电路

Video



- a. Housing
- b. Hot film sensor (fitted in middle of housing)
1. Heat sink
2. Intermediate module
3. Power module
4. Hybrid circuit
5. Sensor element

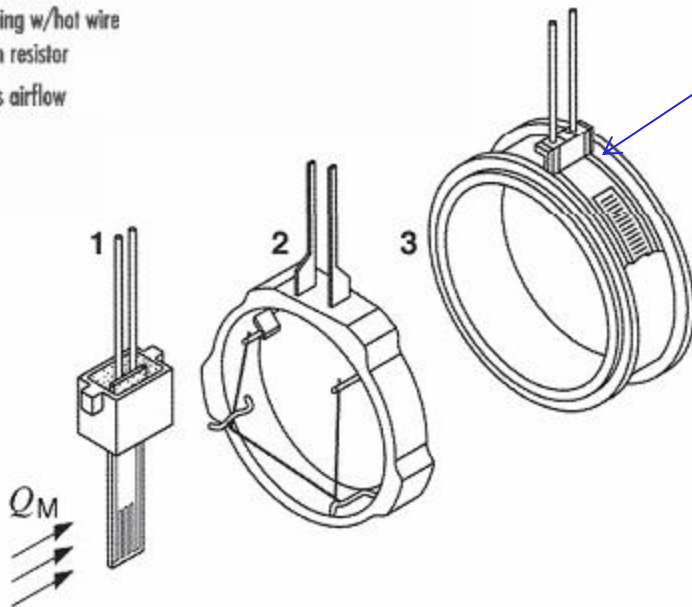


Potentiometer-type

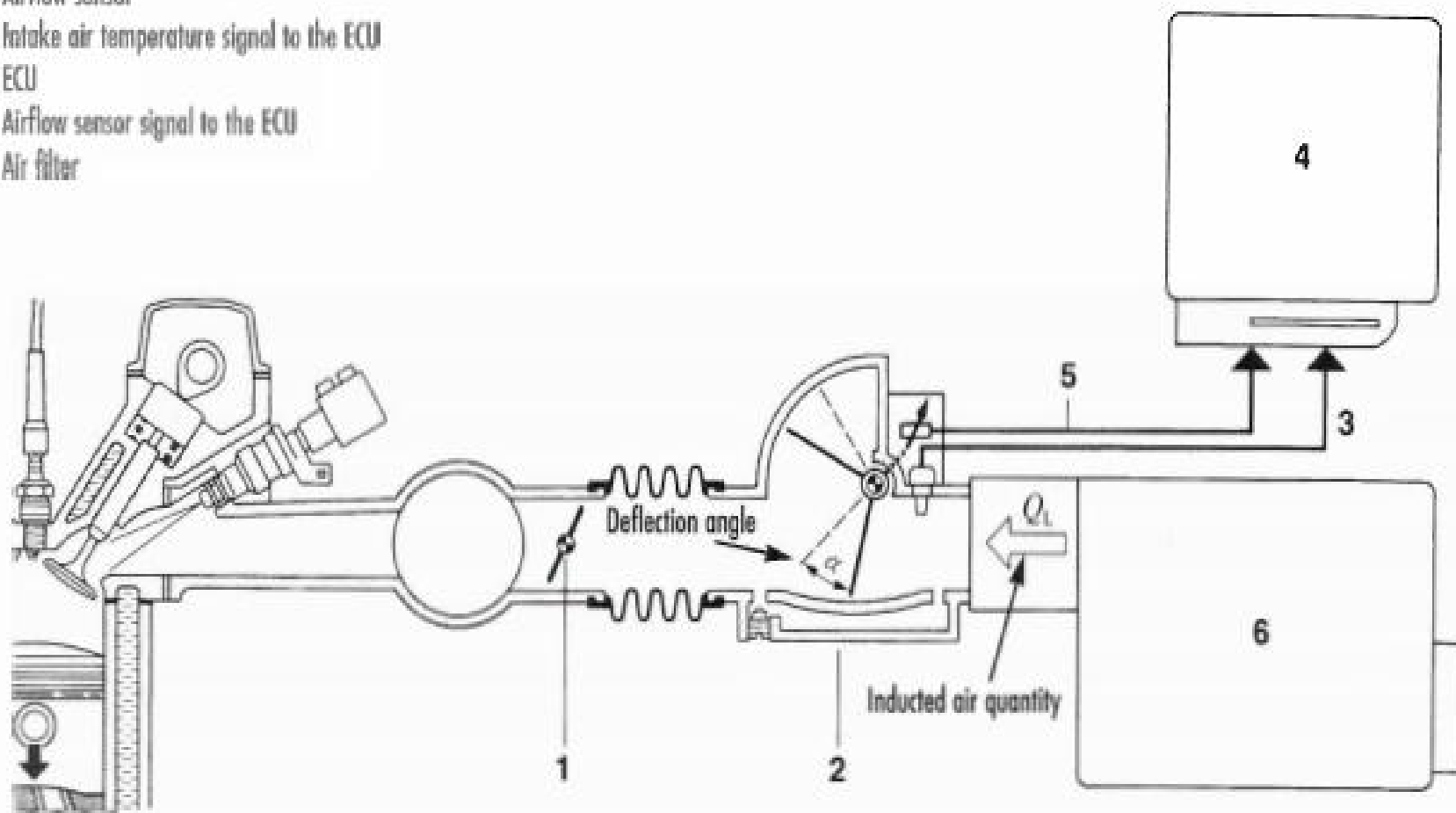
Hot-line type

Hot-film type

1. Temperature sensor
 2. Sensor ring w/hot wire
 3. Precision resistor
- Q_m = Mass airflow



1. Throttle valve
2. Airflow sensor
3. Intake air temperature signal to the ECU
4. ECU
5. Airflow sensor signal to the ECU
6. Air filter



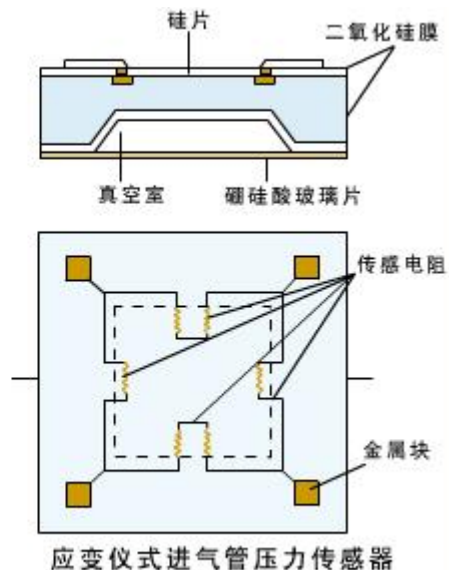
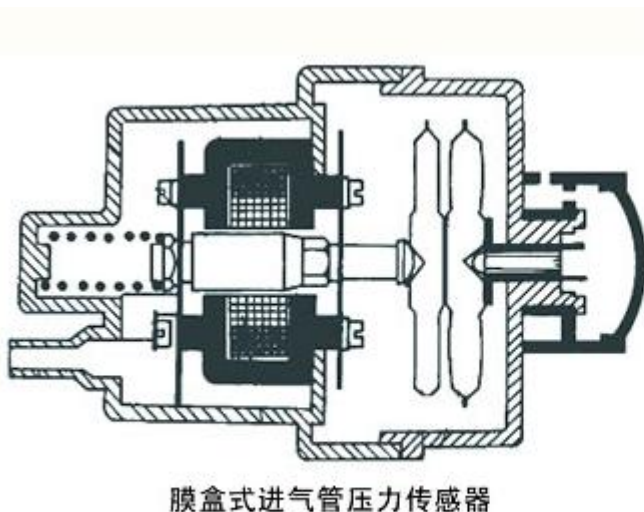
Airflow sensor in the intake system. (Robert Bosch Corporation)

Manifold Air Pressure Sensor

进气管压力(MAP)传感器

在发动机工作时，节气门开大，进气量增多，进气管压力相应增加。因此，进气管压力的大小反映了进气量的多少。

When the engine is working , with the throttle opening , the mass of intake air growing , the manifold air pressure increases at the same time. As a result, the value of the manifold air pressure reflects the amount of intake air.



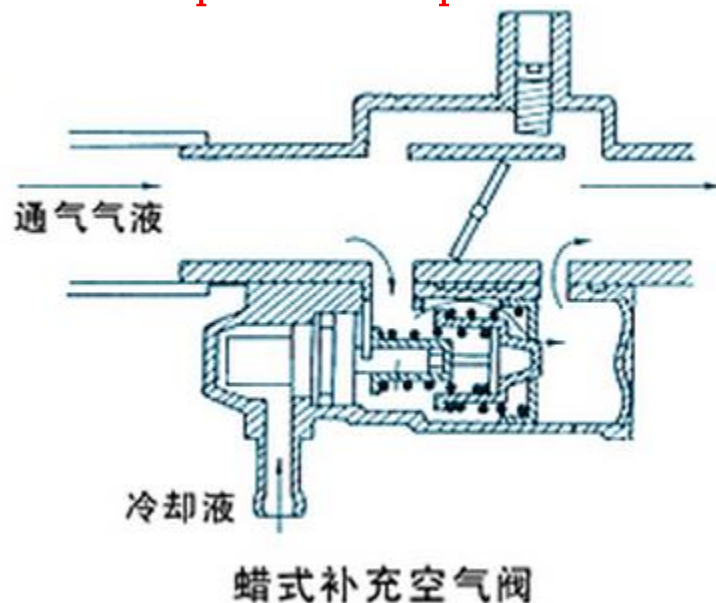
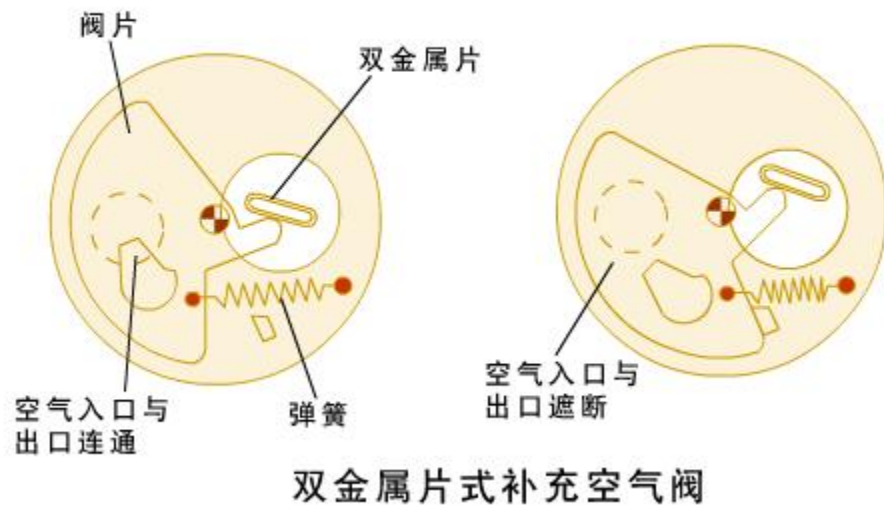
Video

Supplementary Air Valve

补充空气阀

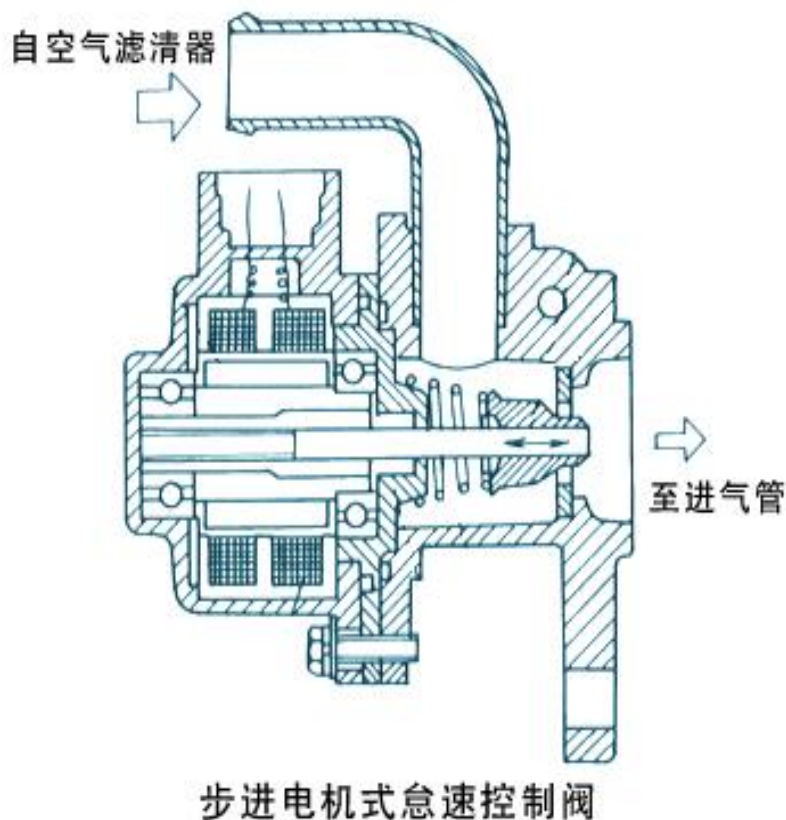
补充空气阀是实现发动机快怠速的装置。当发动机冷起动时，部分空气经补充空气阀进入发动机，使发动机的进气量增加。由于这部分空气是经过空气流量计计量过的，因此喷油量将相应地有所增加，从而提高了怠速转速，缩短了暖车时间。

Supplementary air valve is a fast idle device. With this device, the mass of air drawn in the engine could increase when the engine starts, so as the fuel volume. As a result, the idle speed is improved.



Idle speed control valve

怠速控制阀



在节气门体汽油喷射系统中，节气门体上装有步进电机式怠速控制阀。其功用是自动调节发动机的怠速转速，使发动机在设定的怠速转速下稳定运转。在使用空调器或转向助力器的汽车上，电控单元通过怠速控制阀自动提高怠速转速，以防止发动机因负荷加大而熄火。

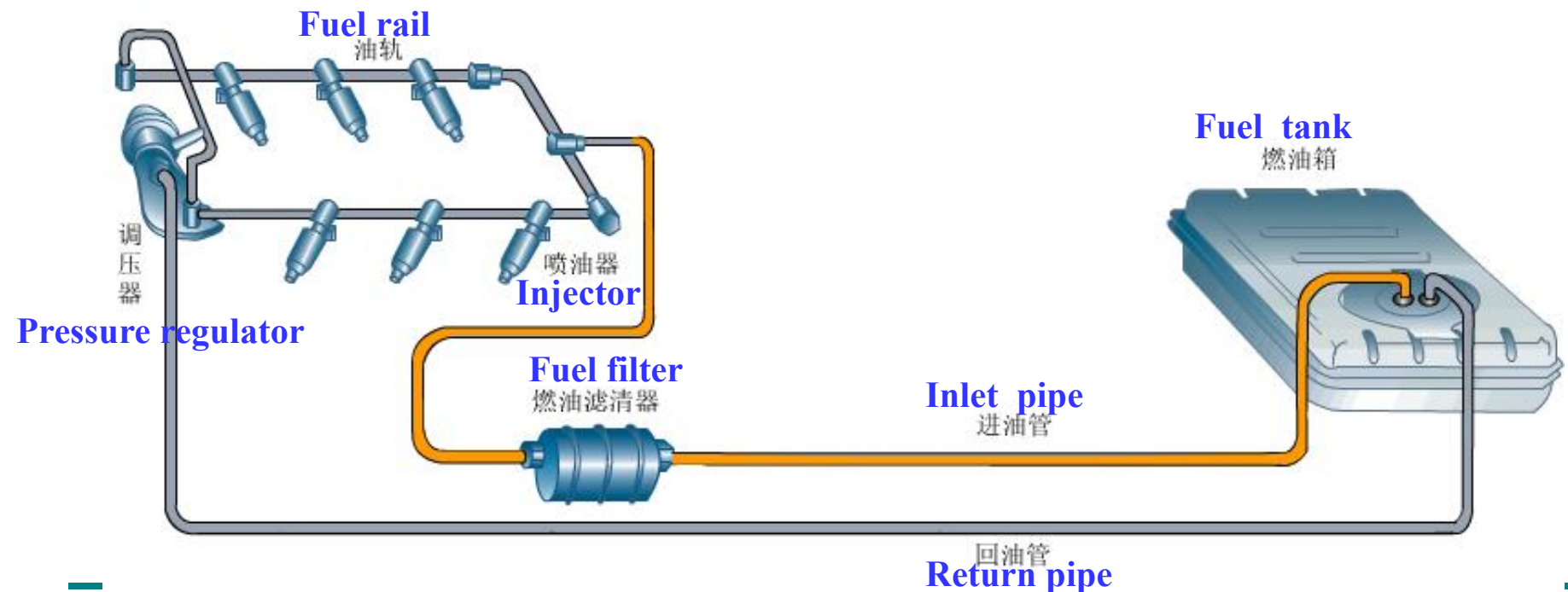
Idle speed control valve is used to modify the idle speed of engines, to ensure the engine to work normally.

Fuel Supply System

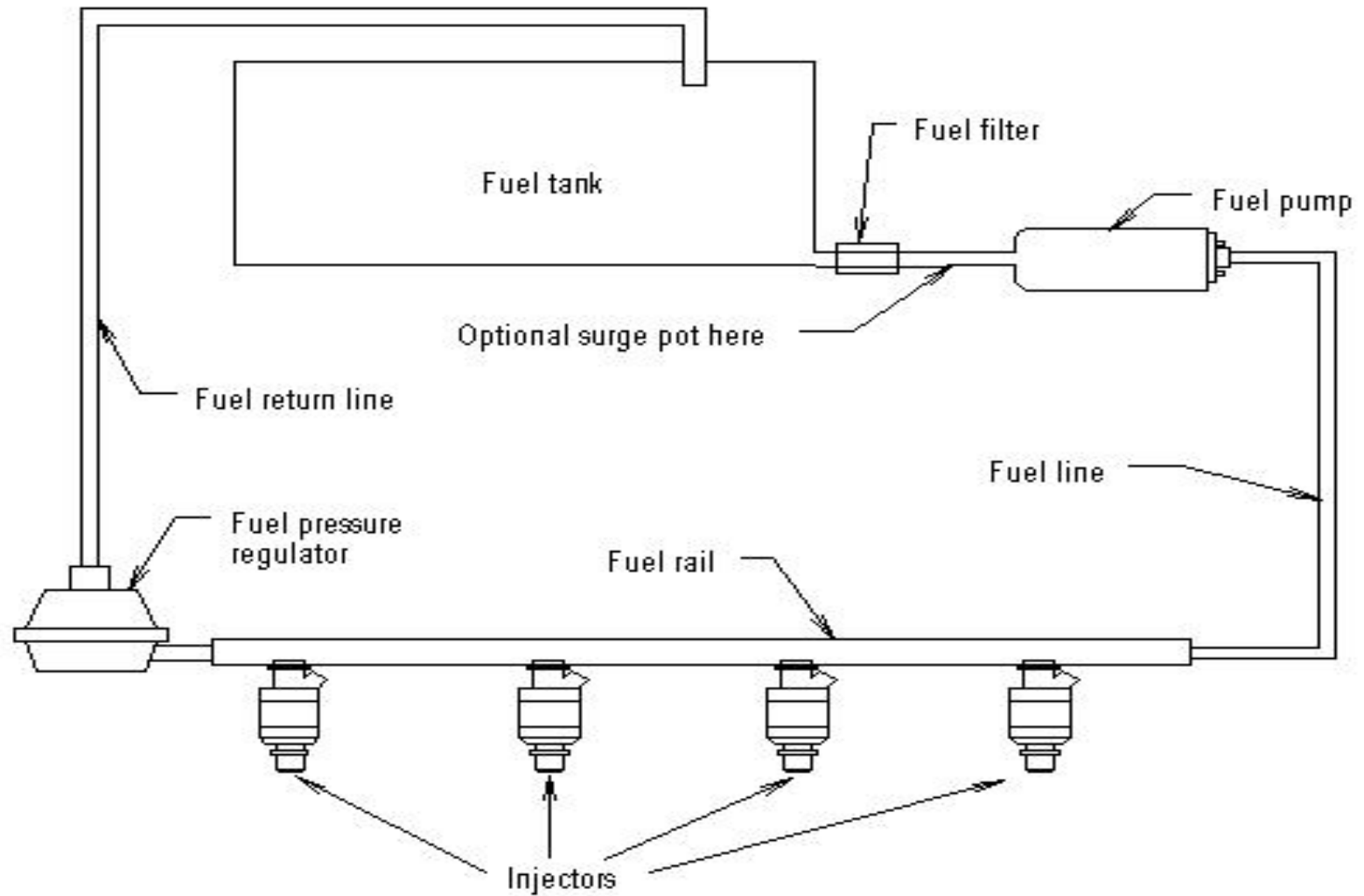
燃油供给系统

燃油供给系统的作用是向发动机供给燃烧所需的汽油。它主要由电动燃油泵、燃油滤清器、燃油压力脉动阻尼器、燃油压力调节器、喷油器和燃油管路等组成。

Fuel supply system mainly consists of fuel pump, filter, pressure pulsation damper, pressure regular, injector and fuel rail.



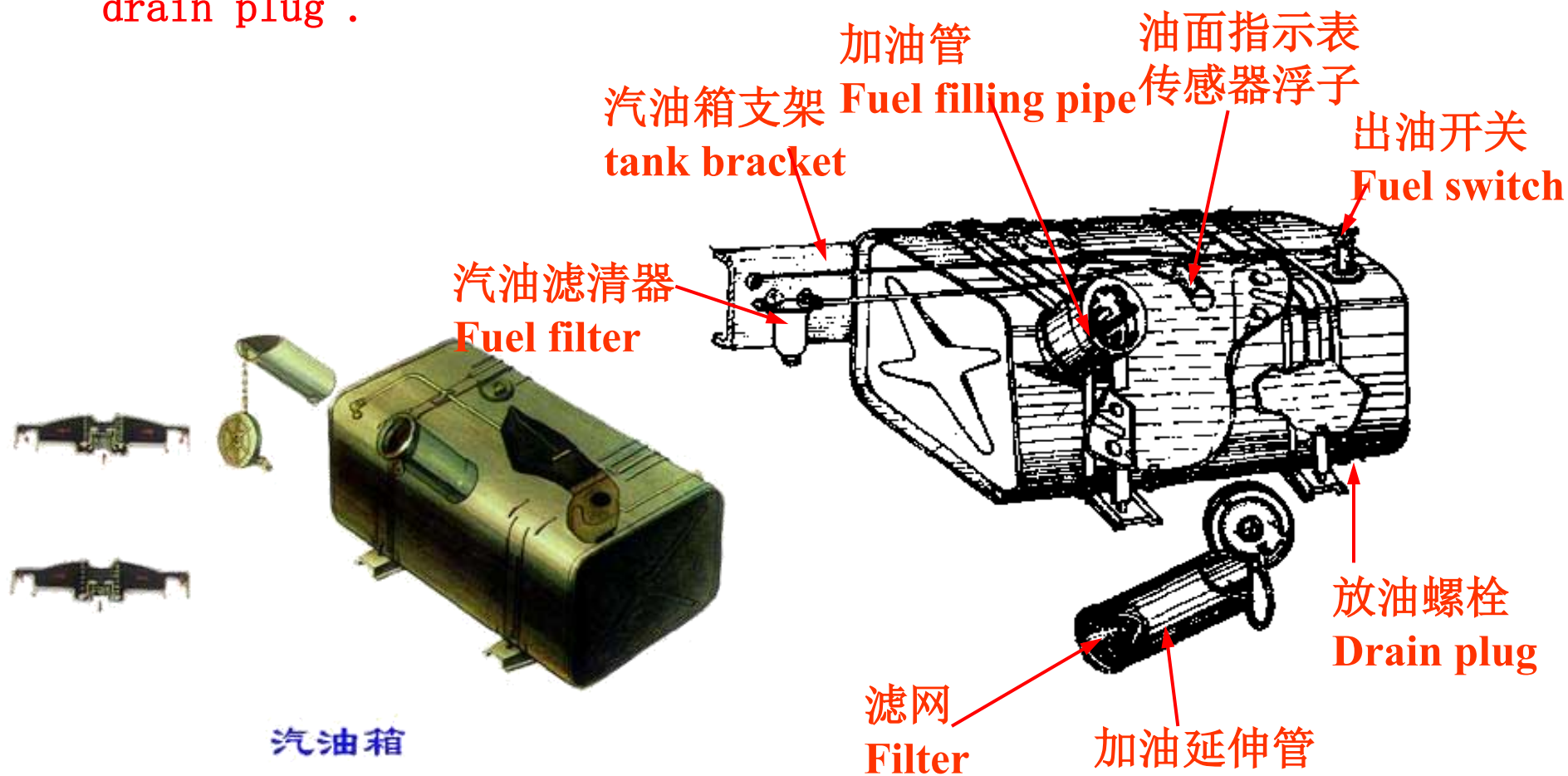
Basic injection system



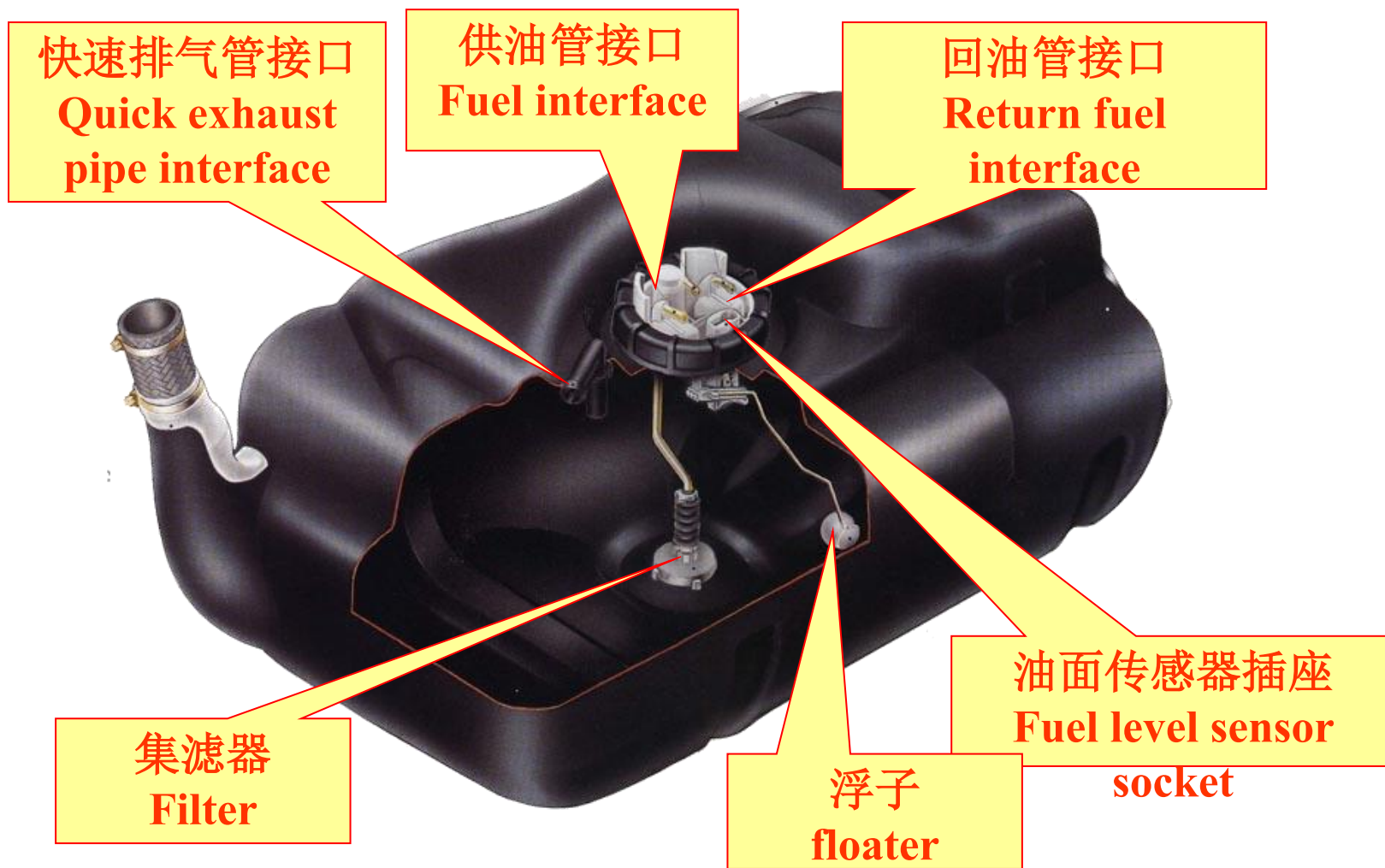
[Video](#)

Fuel Tank汽油箱

汽油箱的功用是储存汽油。在汽油箱上还装有油面指示表传感器、出油开关和放油螺塞等。The fuel tank is also equipped with the fuel level instruction sensor , fuel switch and the fuel drain plug .



桑塔纳轿车汽油箱



Fuel tank cap 汽油箱盖

- 作用：密封汽油箱。

Used to seal the fuel tank

Steam valve

蒸汽阀

弹簧

spring

空气阀

Air valve

弹簧

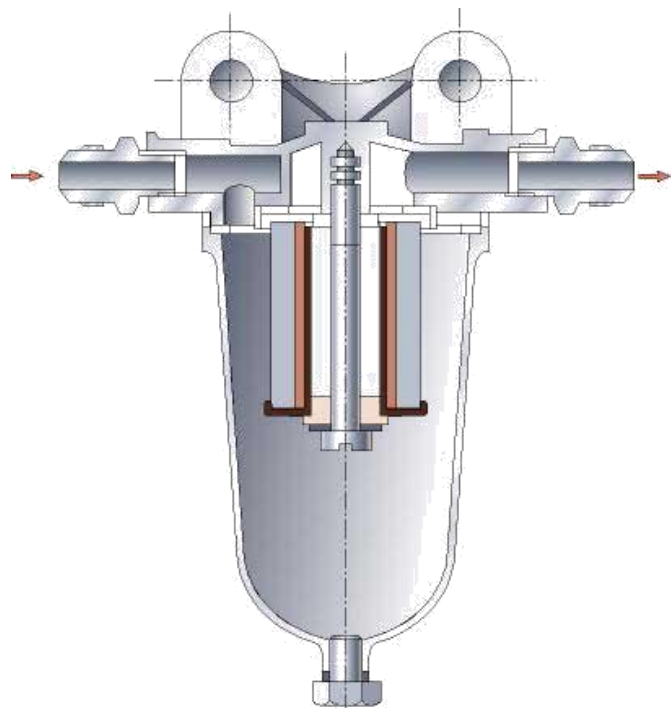
spring



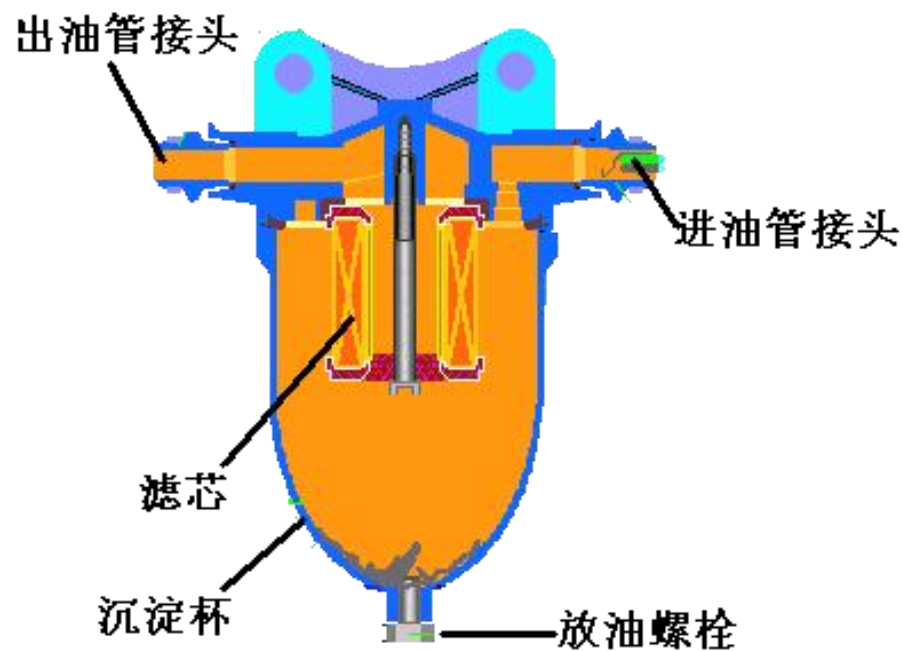
working process of the cap 汽油箱盖的工作过程

Fuel filter 汽油滤清器

- 功用：除去汽油中的水分和杂质，使汽油能达到发动机工作的需要。
- To remove the moisture and impurities in the gasoline, and fulfill the working requirement of engine.



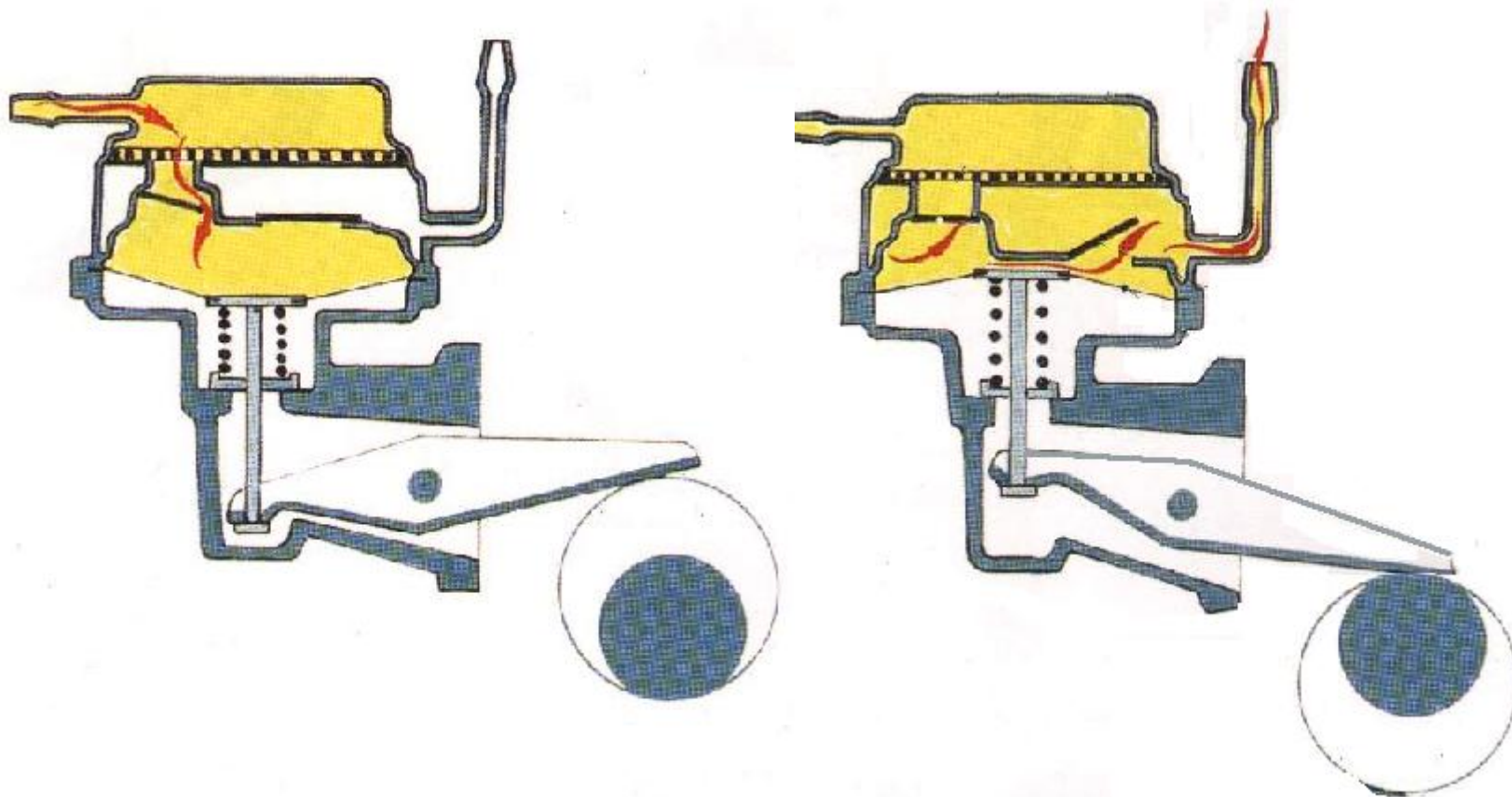
汽油滤清器



进、出油口不可装反

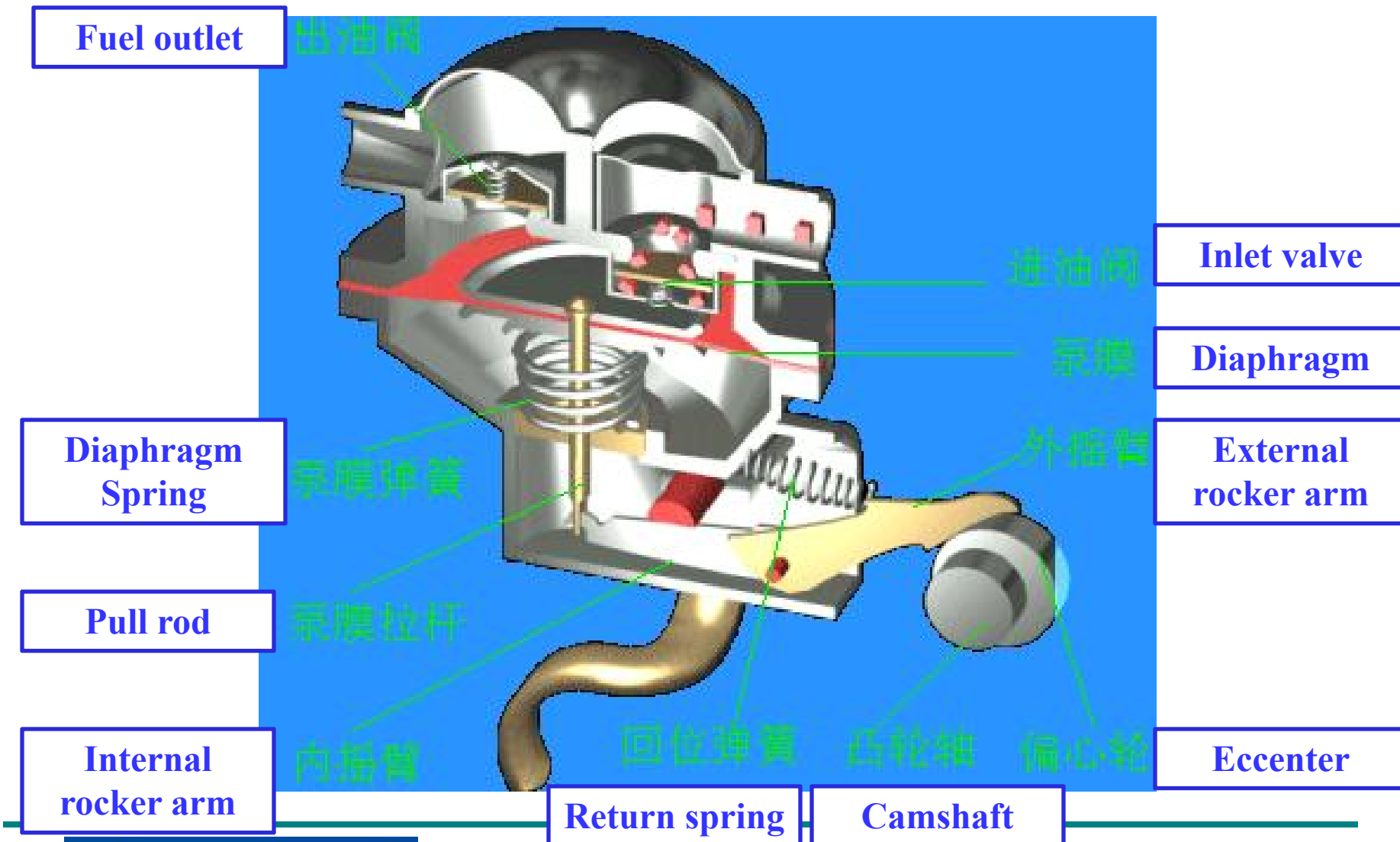
Fuel Pump 汽油泵

- 功用：将汽油从油箱中吸出，经管路和汽油滤清器，然后泵入化油器浮子式。
- Used to suck the fuel out of fuel tank.



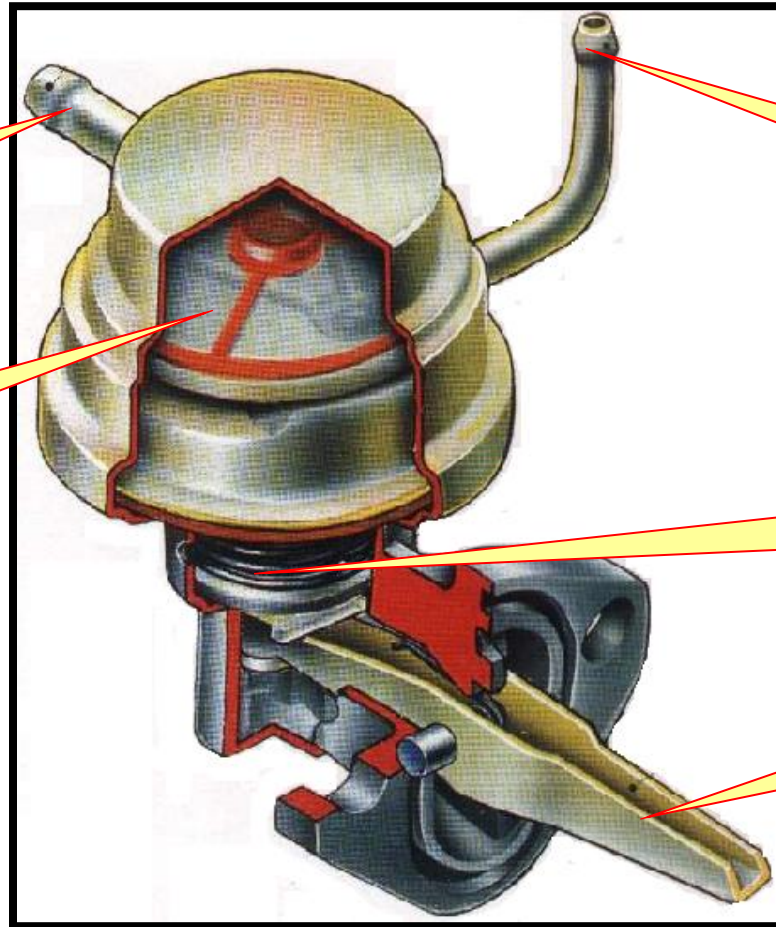
Mechanical fuel pump

机械式汽油泵工作原理演示



Fuel pump for SANTANA

桑塔纳发动机汽油泵



进油口
Fuel inlet

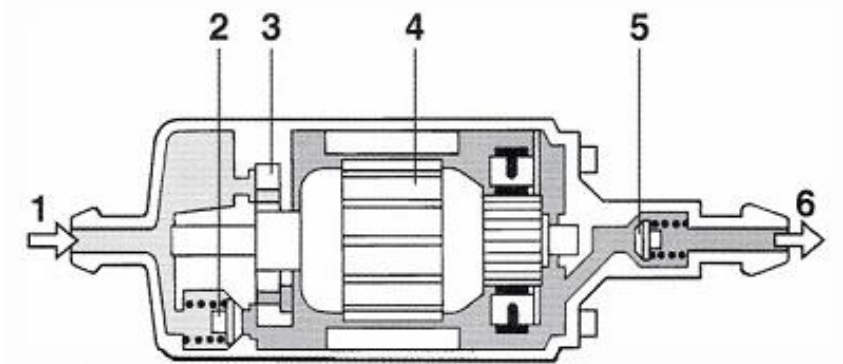
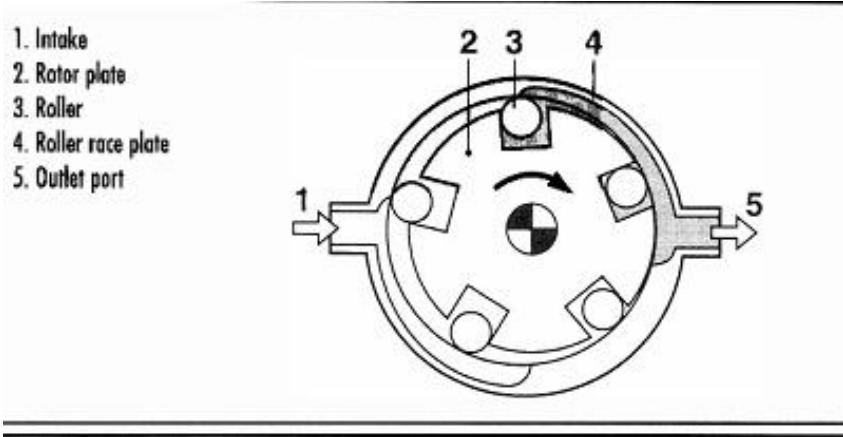
滤网
Filter

出油口
Fuel outlet

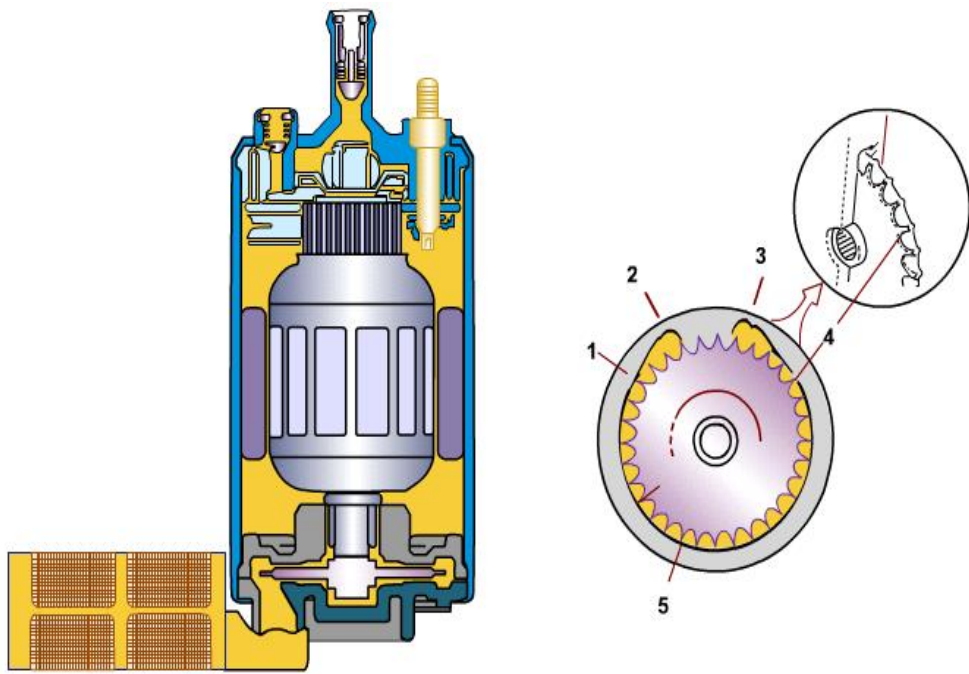
回位弹簧
Return spring

摇臂
Rocker arm

滚柱式电动燃油泵 Roller cell fuel pump



涡轮式电动燃油泵 Turbine fuel pump



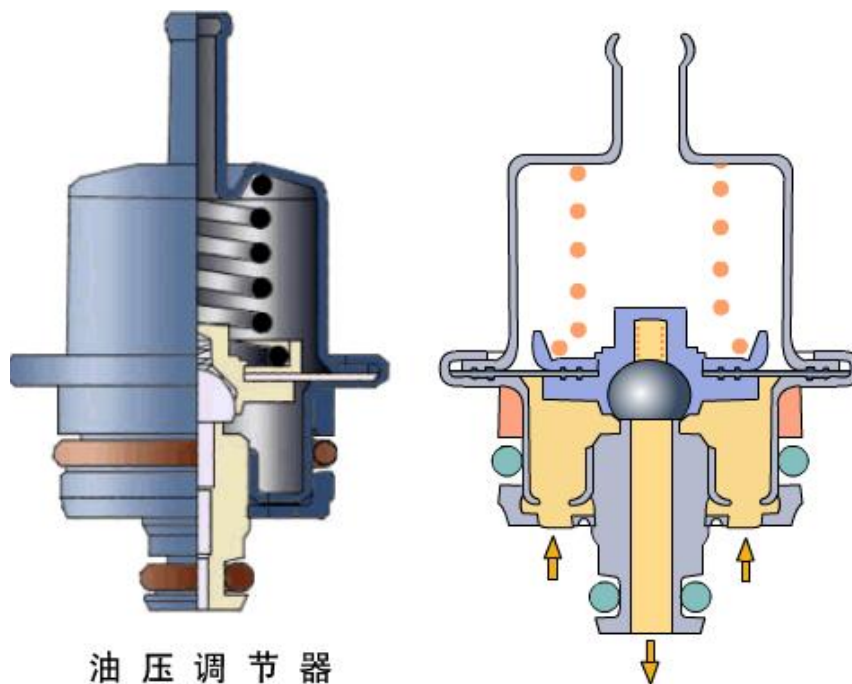
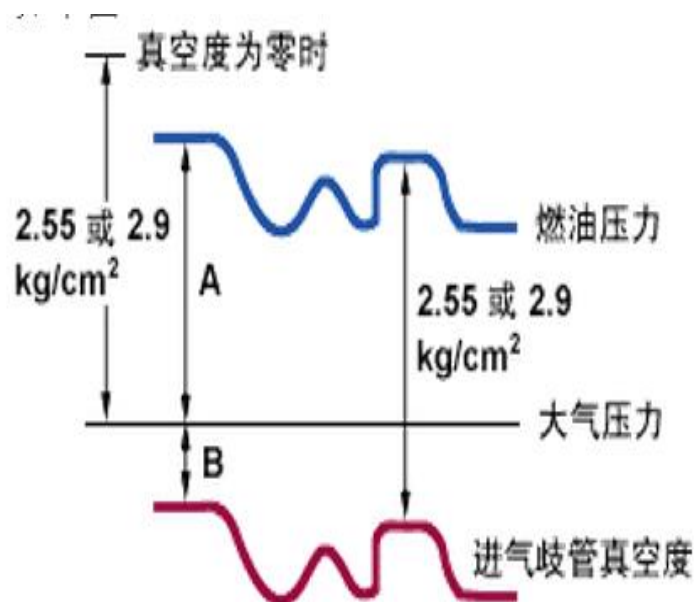
Fuel pressure regulator 油压调节器

油压调节器的功用是使燃油供给系统的压力与进气管压力之差即喷油压力保持恒定。

Used to keep the injection pressure invariable.

喷油压力 = 燃油压力 - 进气歧管压力

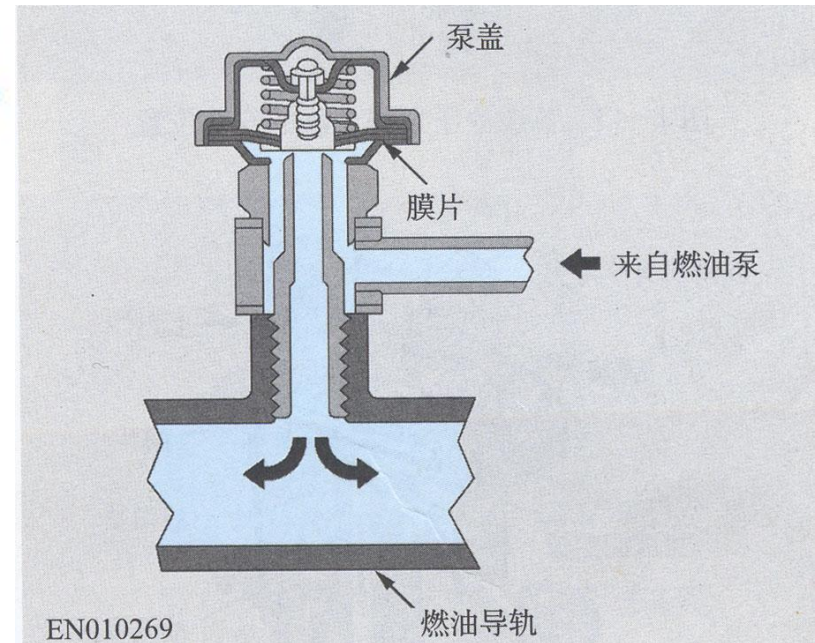
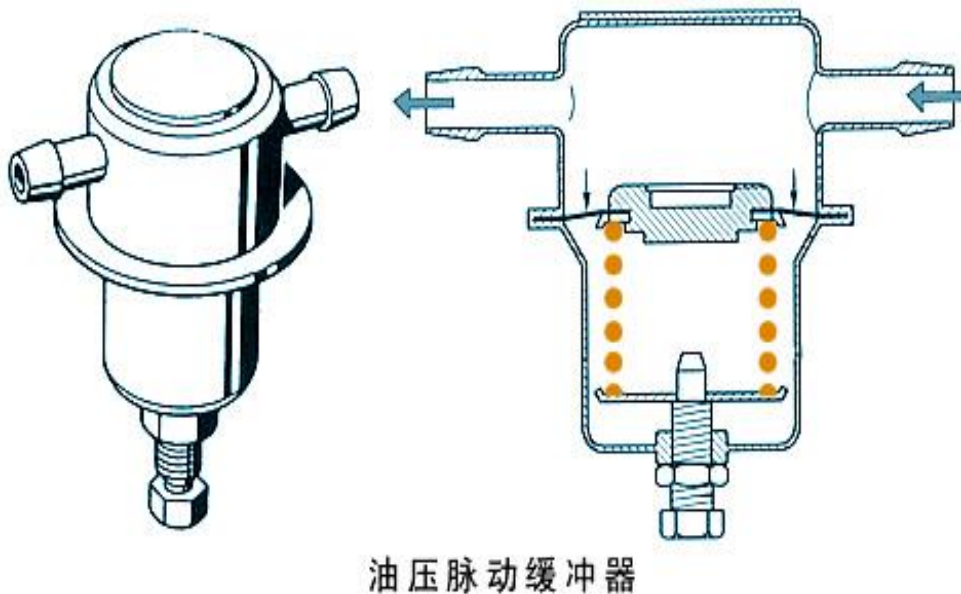
Injection pressure = fuel pressure - manifold pressure



Pressure pulsation damper 油压脉动缓冲器

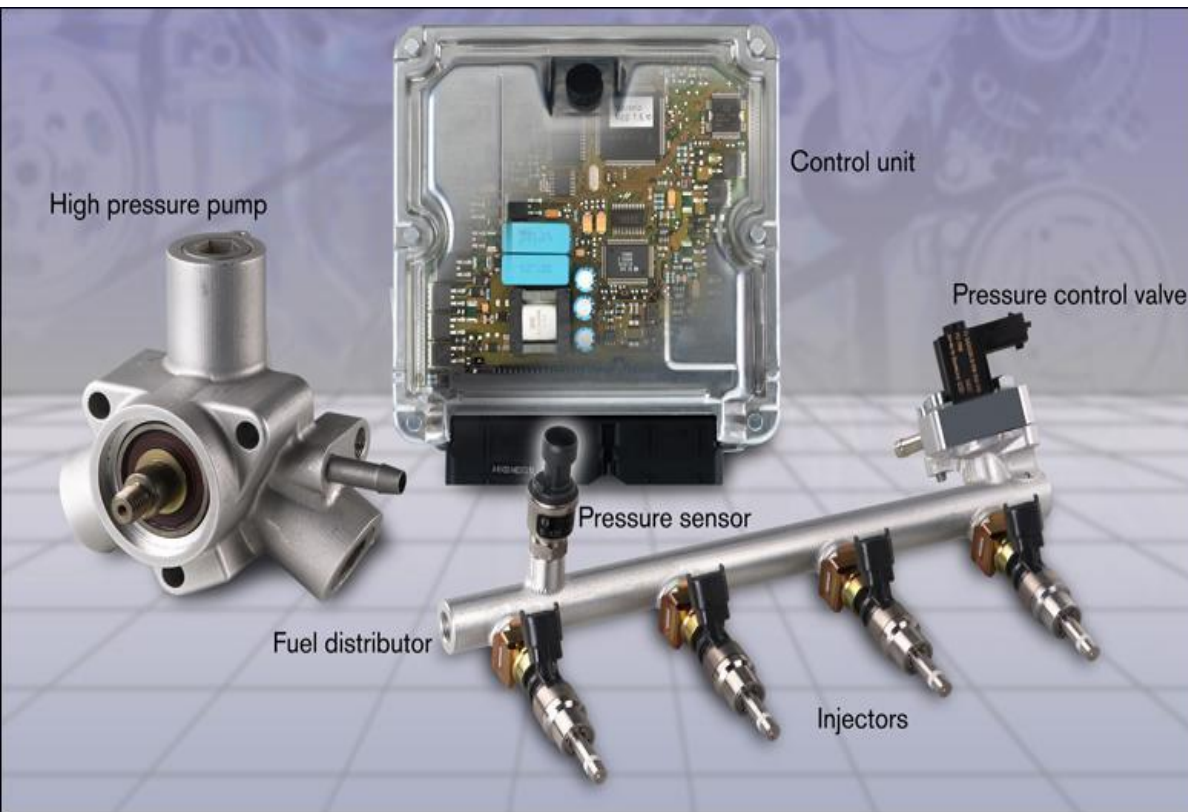
油压脉动缓冲器的作用就是减小燃油管路中油压的脉动和脉动噪声，并能在发动机停机后保持油路中有一定的压力，以利于发动机重新启动。

Used to reduce the pressure pulsations and pulsation noise in the fuel pipe, and can maintain a certain pressure after the engine stop working, so the engine is easy to restart.

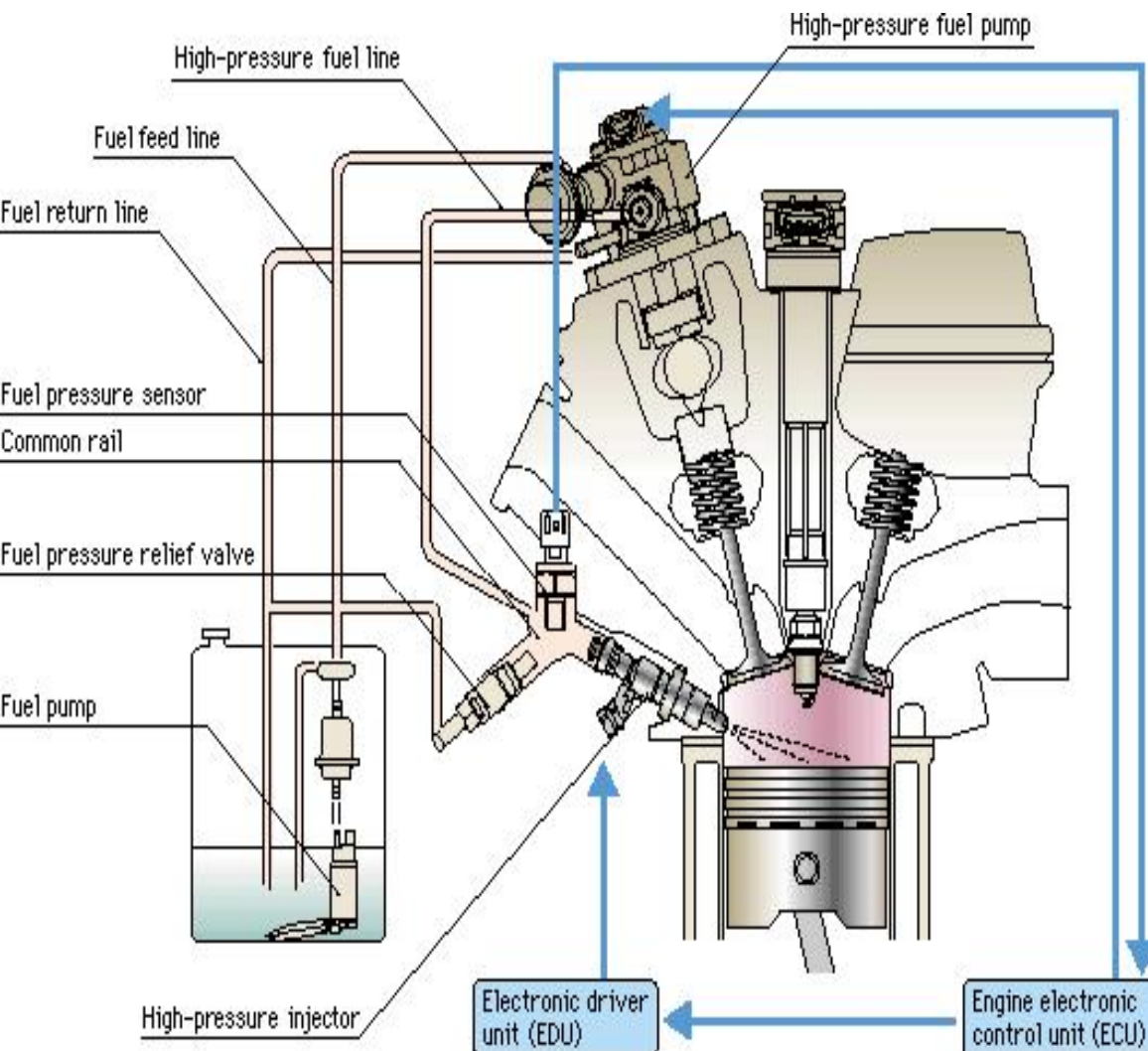


Pressure regulation for high pressure injection

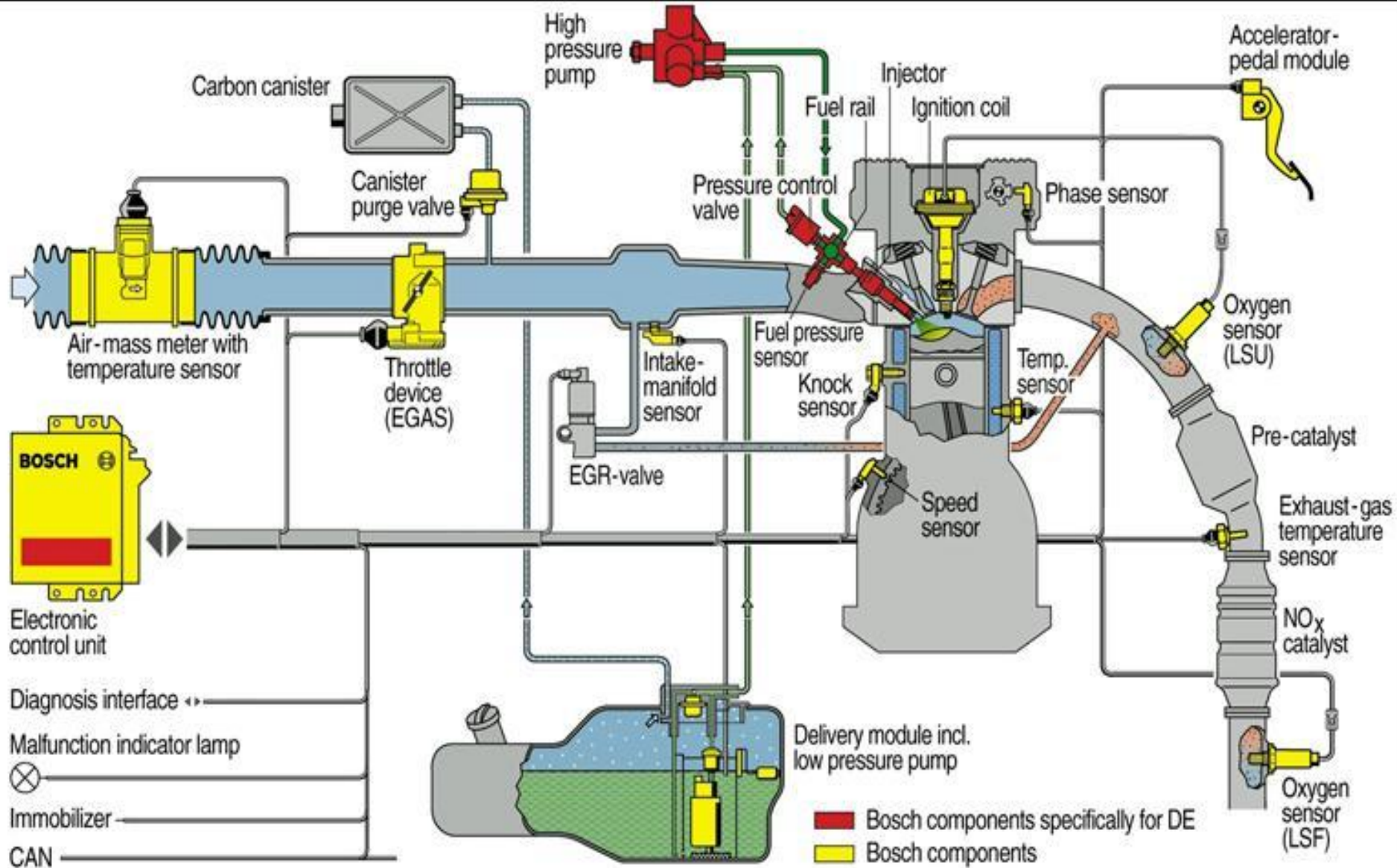
In direct gasoline-injection engines, the A/F mixture is formed directly in the combustion chamber. During the intake stroke, the fuel is injected at high pressure into the combustion chamber .



Different with the low pressure injection, the high pressure injection system is equipped with a high pressure pump. Fuel from the low pressure pump is compressed and transmitted into common rail. The pressure is regulated by ECU, pressure sensor and pressure control valve.



Fuel pressure sensor mounted on the fuel rail selects the pressure valves and transmits them to ECU. ECU decides whether the current pressure meet requirement of engine. If the pressure is low, high pressure pump will keep increasing it. If the pressure is higher than requirement, ECU makes the pressure control valve open, and fuel flows back to tank from rail.

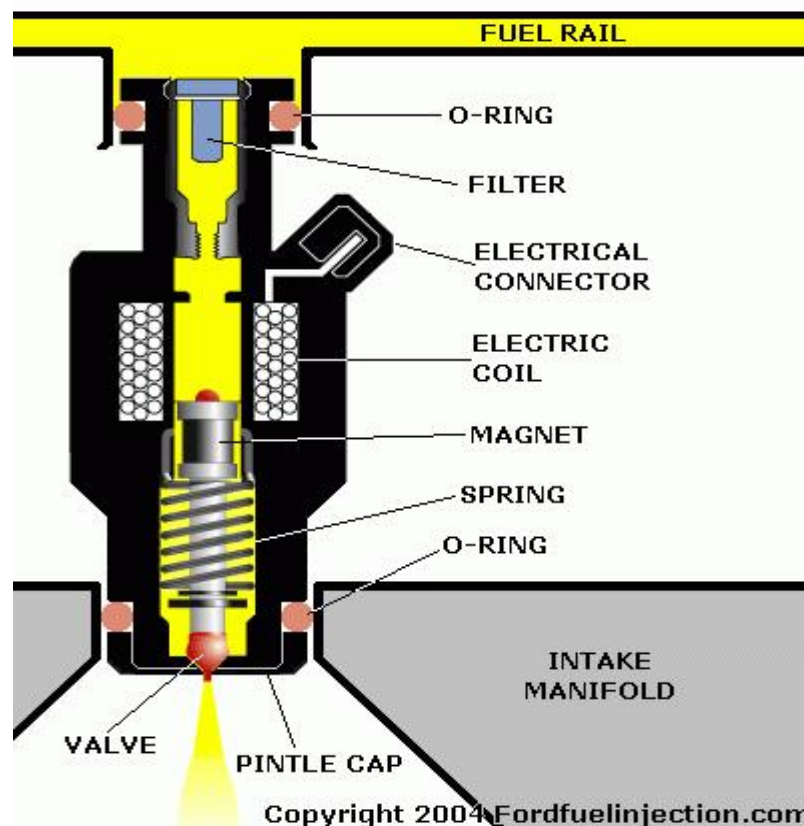
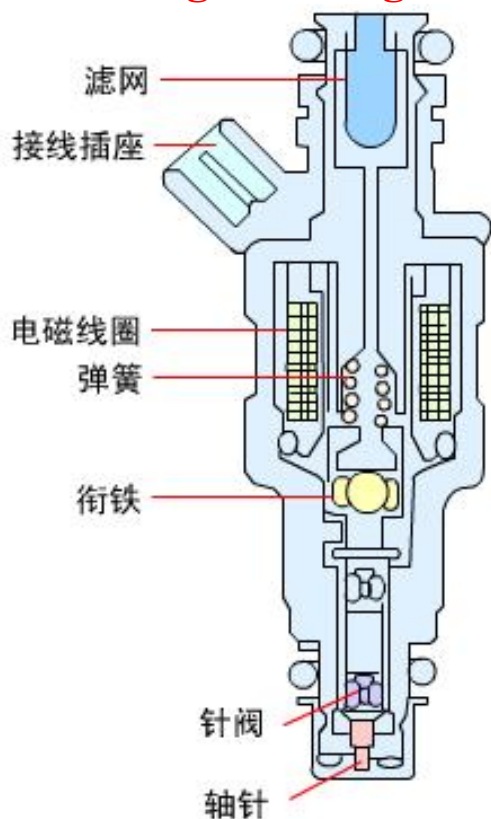


Typical G-DI engine system layout

Injector 喷油器

喷油器的功用是按照电控单元的指令将一定数量的汽油适时地喷入进气道或进气管内，并与其中的空气混合形成可燃混合气。

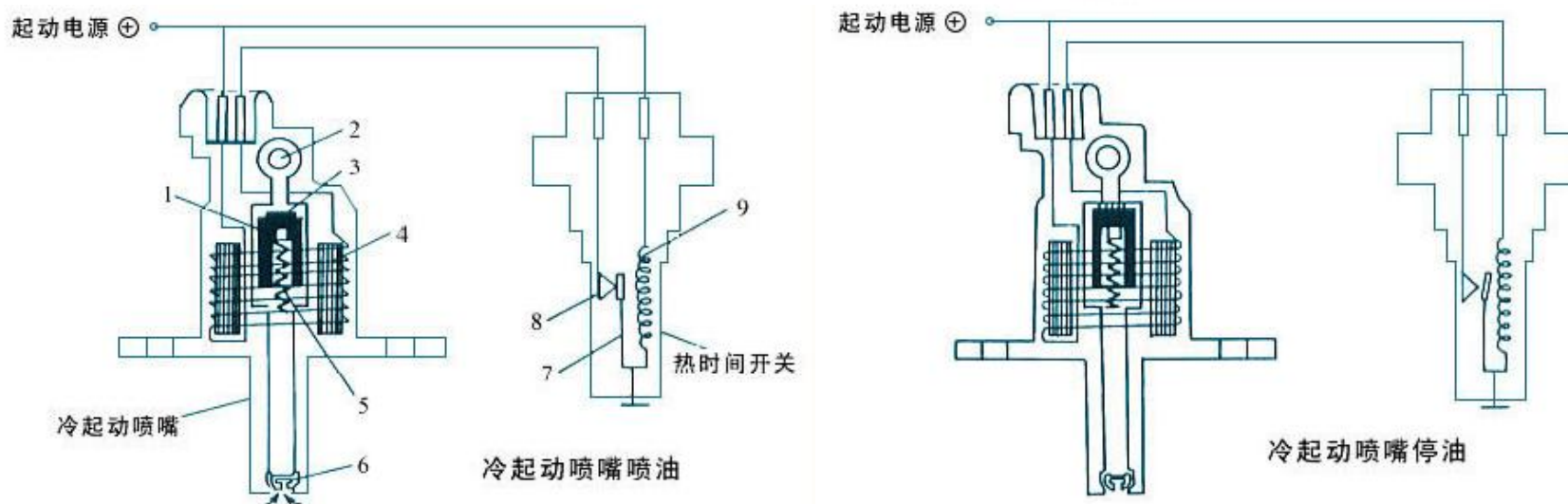
The function of injector is injecting a certain amount of gasoline into manifold timely according to the signal from ECU.



Cold start injector 冷起动喷嘴

冷起动喷嘴的功用是当发动机低温起动时，向进气管喷入一定数量附加的汽油，以加浓混合气。

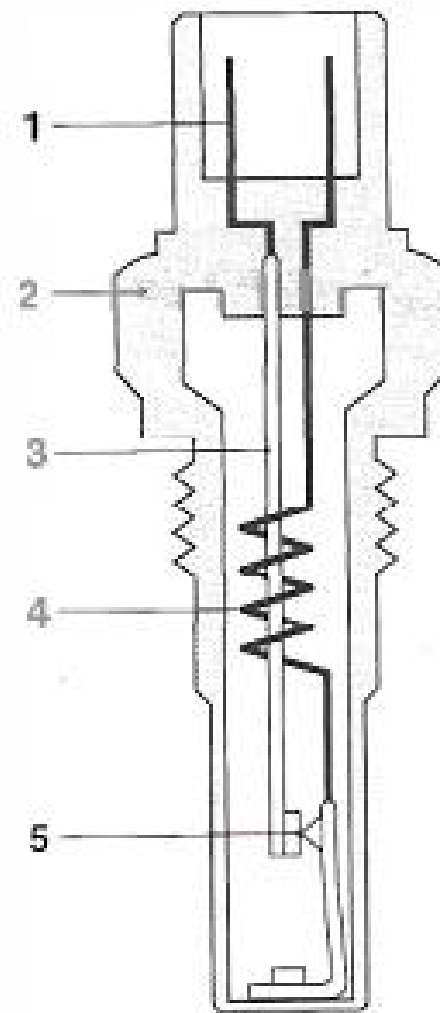
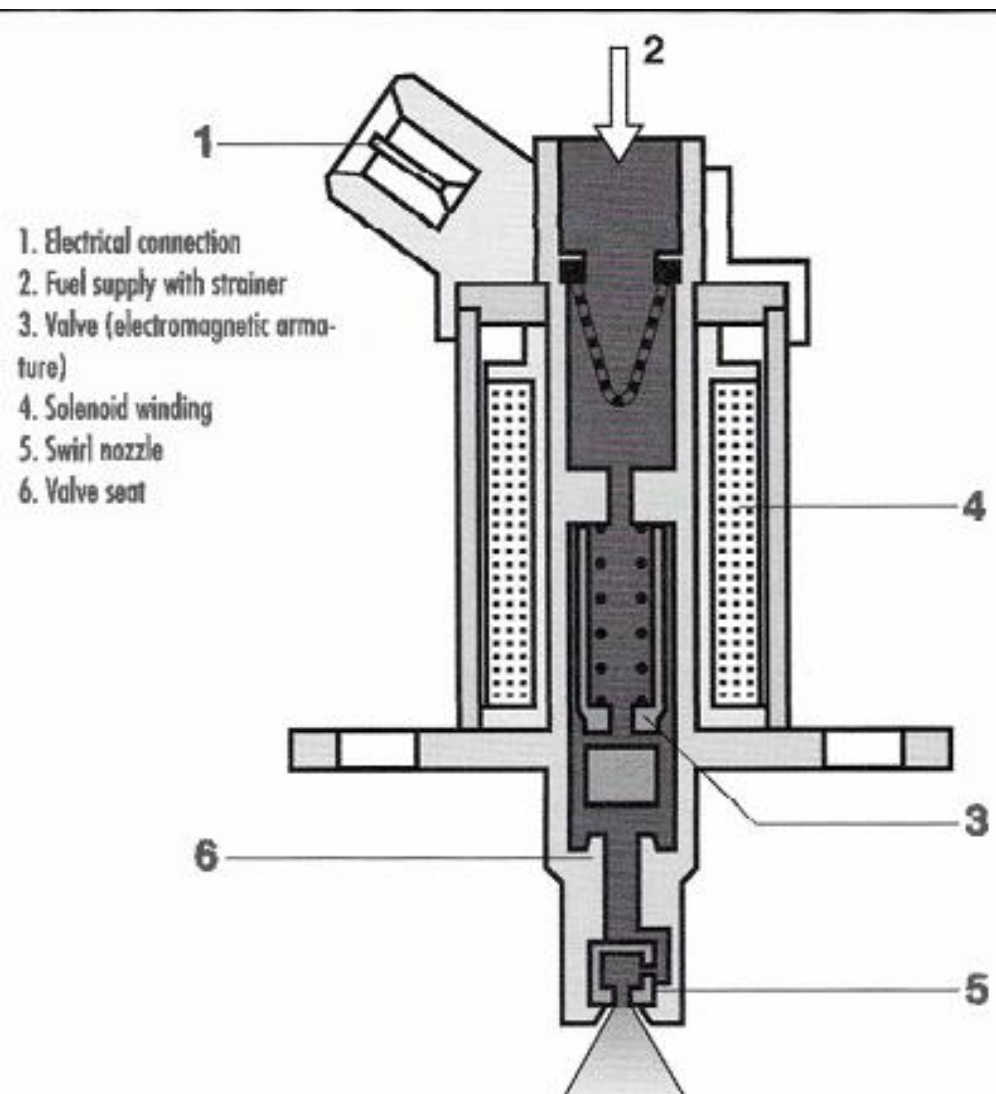
Used to inject a certain amount of extra gasoline to enrich the mixture when the engine cold starts.



冷起动喷嘴和热时间开关及其工作原理

1-可动铁心；2-进油口；3-平面阀；4-电磁线圈；5-弹簧；6-涡流式喷嘴；
7-双金属片；8-白金触电；9-电热丝

Cold start injector and thermo time-switch



Electronically Controlling System

电子控制系统

传感器、执行元件和电子控制单元三部分组成。

Electronically controlling system consists of different sensors, power element and electronic control unit(ECU).

电子控制系统的功用是根据各种传感器的信号，由计算机ECU进行综合分析和处理，通过执行装置控制喷油量等，使发动机具有最佳性能。

The function of the electronically controlling system is to rely on the ECU comprehensively analyzing and calculating the signals from different kinds of sensors, and control the mass of fuel injected into manifold by the power element. As a result , the engine could work at best condition.



Sensor 传感器

Sensors are used to select signals of the engine working condition, and to transmit these signals to ECU.



氧传感器
Lambda sensor

Load sensor
负荷传感器



转速传感器
Rotation speed sensor



温度传感器
Temperature sensor



节气门传感器
Throttle position sensor



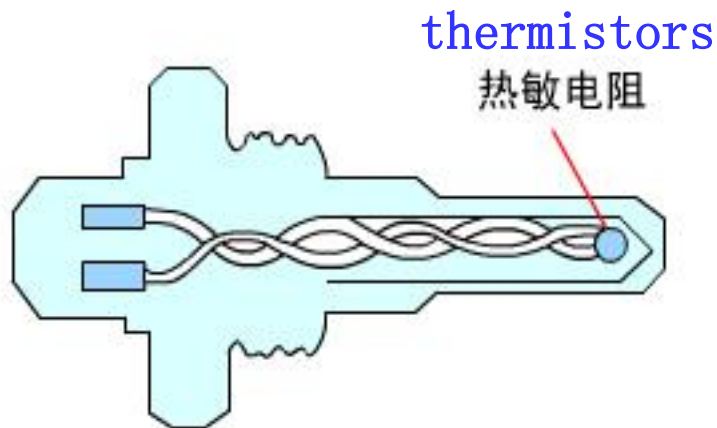
爆震传感器
Knock sensor

Temperature Sensors 温度传感器

包括发动机温度传感器（水温传感器）和进气温度传感器。

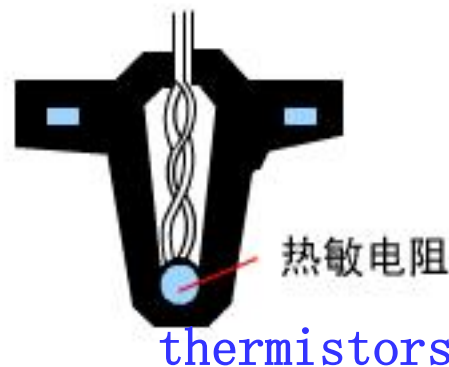
Include coolant temperature sensor and intake temperature sensor.

负特性 **negative characteristic**



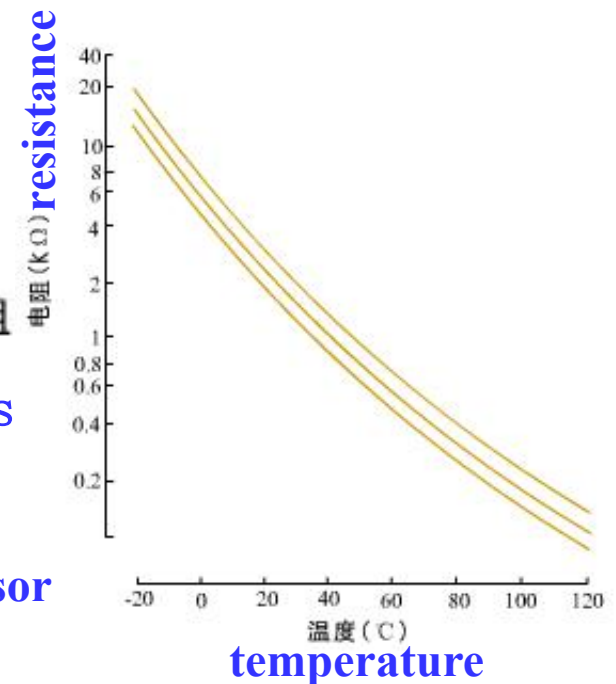
(a) 水温传感器

Coolant temperature sensor



(b) 进气温度传感器

Intake temperature sensor

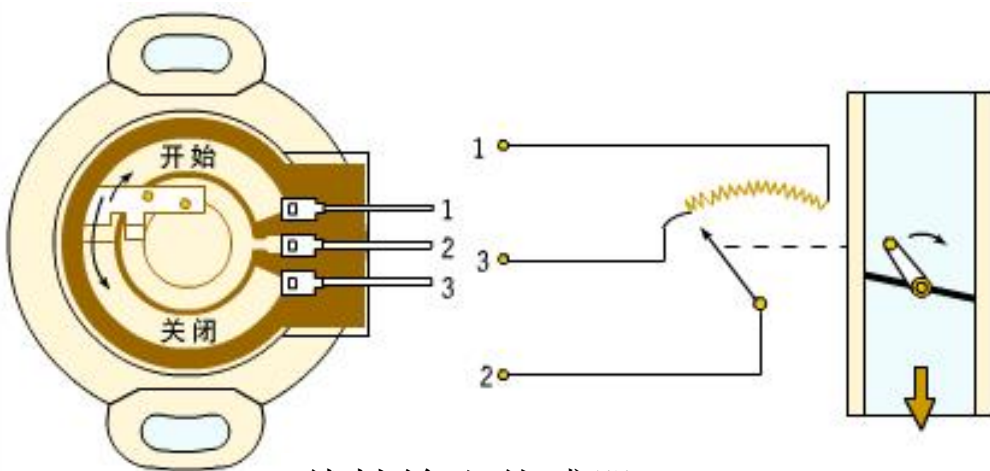


Throttle position sensor (TPS)节气门位置传感器

其功用是将节气门的位置或开度转换成电信号传输给电控单元，作为电控单元判定发动机运行工况的依据。节气门位置传感器有开关型和线性输出型两种。

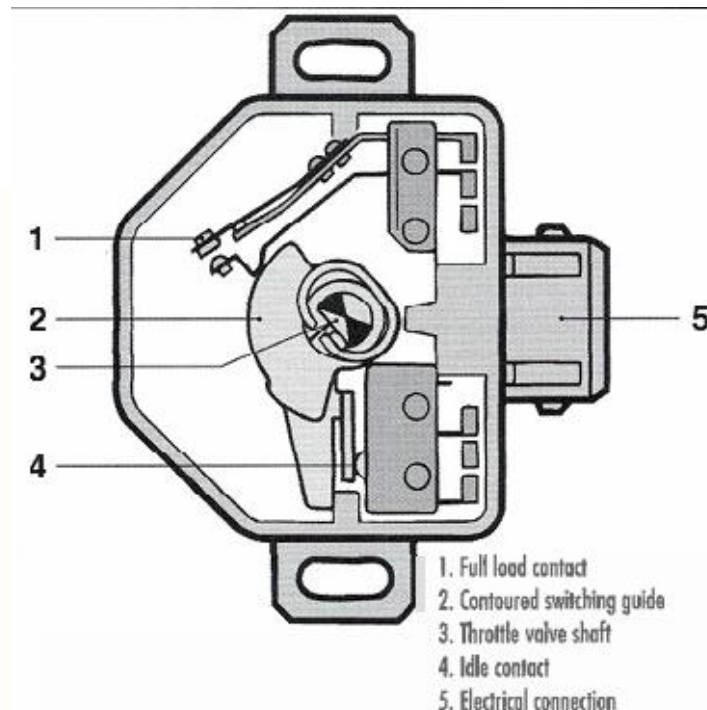
It's used to change the position of throttle into electrical signals and transmit them to ECU as the basis of deciding working condition of engine. There are two kinds:

switch type / linear output type



线性输出传感器

linear output type



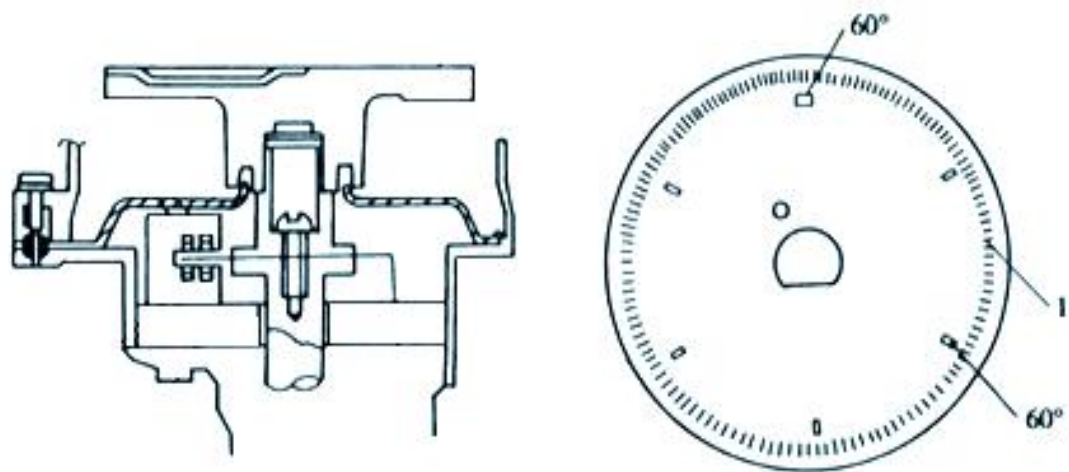
switch type

Crankshaft Position Sensor 曲轴位置传感器

用来检测发动机转速、曲轴转角以及作为控制点火和喷射信号源的第一缸和各缸压缩行程上止点信号。

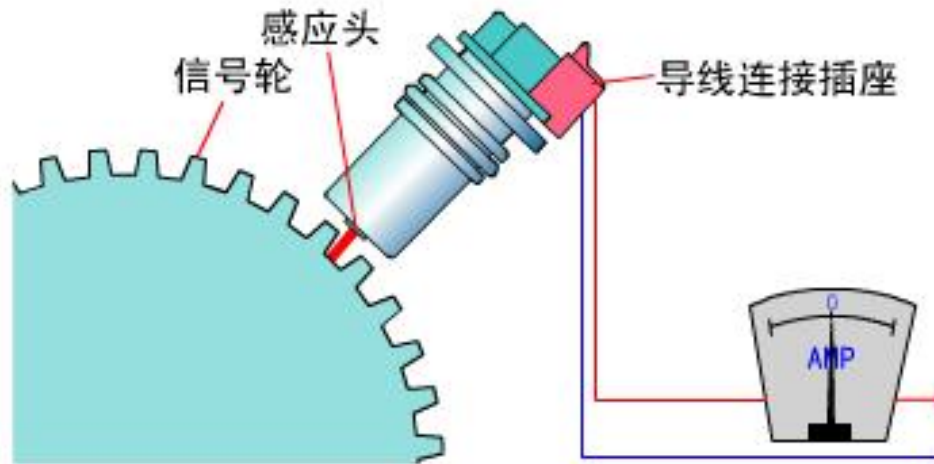
Used to detect the engine speed, crankshaft angle and top dead centre to control ignition and injection.

光电式曲轴位置传感器 Photoelectric type CPS

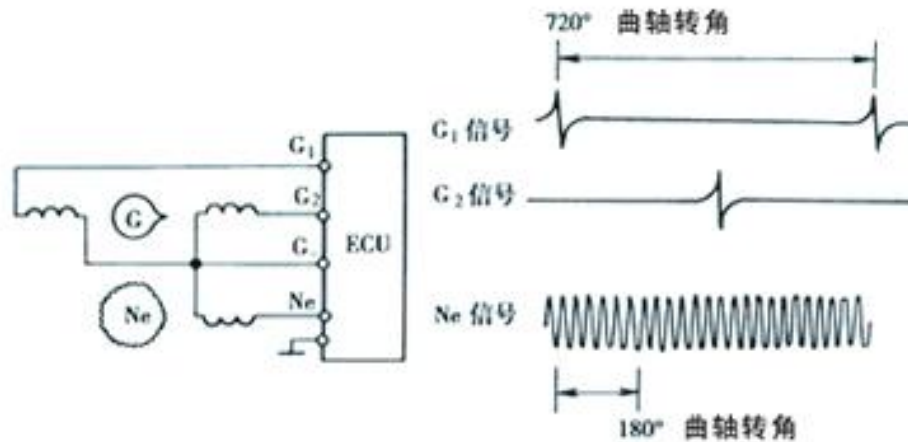
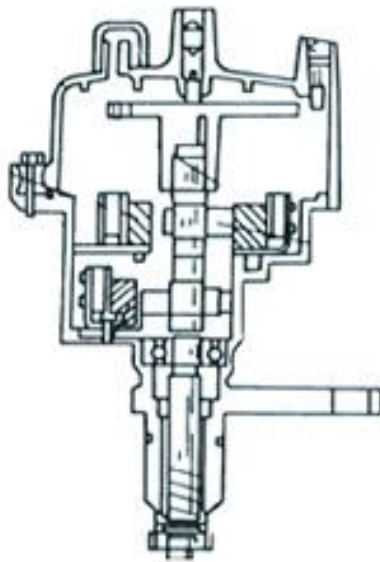


光电式曲轴位置传感器

Magnetic pulse type CPS 磁脉冲式曲轴位置传感器



Video

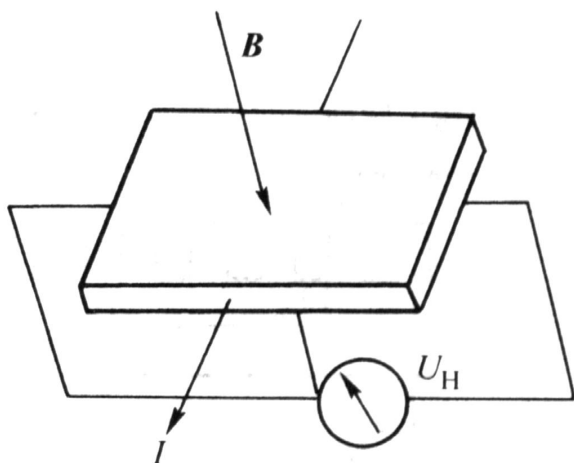


磁脉冲式曲轴位置传感器

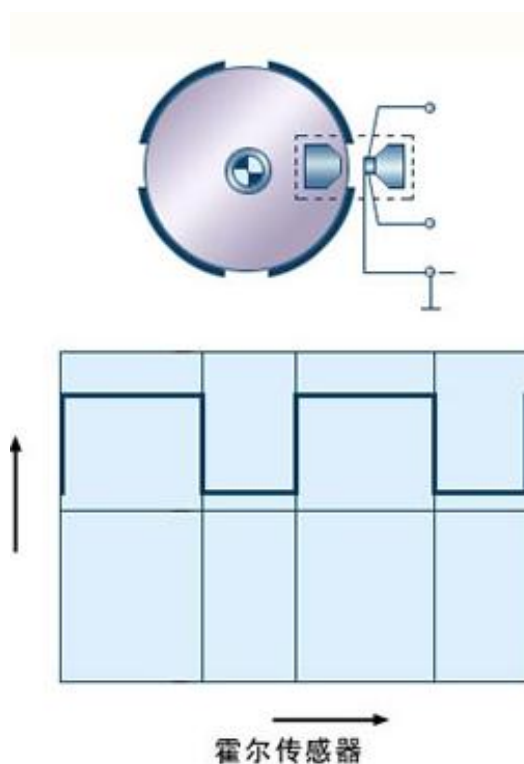
Hall effect type CPS 霍尔效应式曲轴位置传感器

它是利用霍尔效应原理，产生与曲轴转角相对应的电压脉冲信号进行工作的。

It take use of the principle of the hall effect to generate the voltage pulse signal corresponding to the the crankshaft angle.



- $U_H = R_H I B / d$
- R_H ——霍尔系数
- I ——电流强度
- B ——磁场强度
- d ——基片厚度

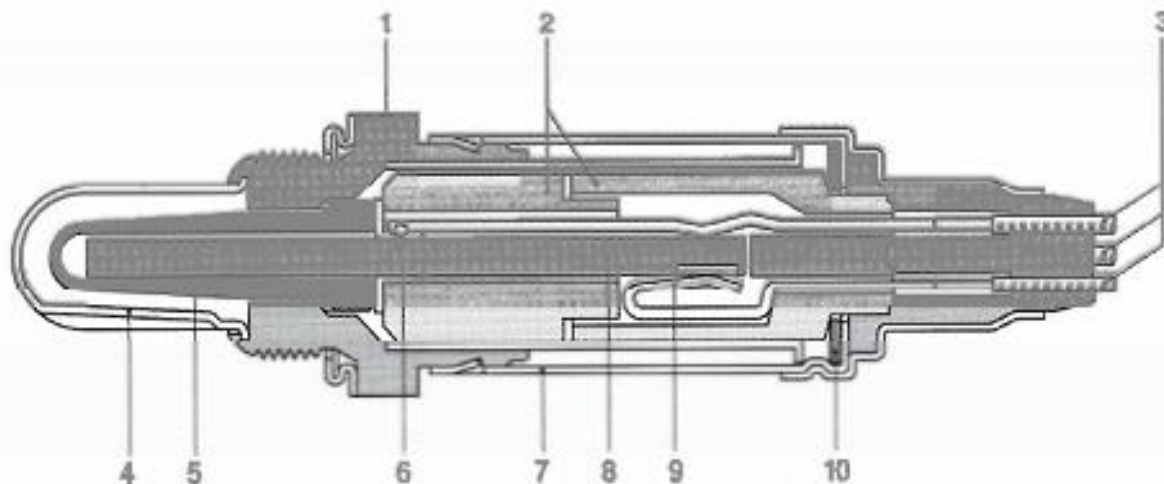


Lambda (Oxygen) sensor 氧传感器

氧传感器是电子控制汽油喷射系统进行反馈控制的传感器，安装在排气管上，电控单元必须根据氧传感器的反馈信号，控制混合气的空燃比更接近于理论空燃比。目前应用最多的是氧化锆氧传感器。

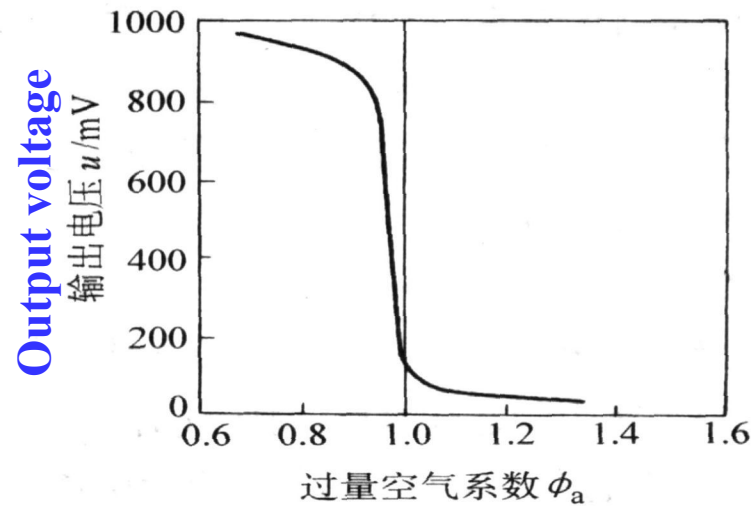
Lambda sensor is used for feedback control in the injection system, mounted on the exhaust pipe. According to the signal from lambda sensor, ECU could control the air-fuel ratio more close to the stoichiometric ratio. The zirconia oxygen sensor is widely used.

1. Probe housing
2. Ceramic shield
3. Electrical connections
4. Shield tube w/slits
5. Active ceramic sensor layer
6. Contact
7. Shield

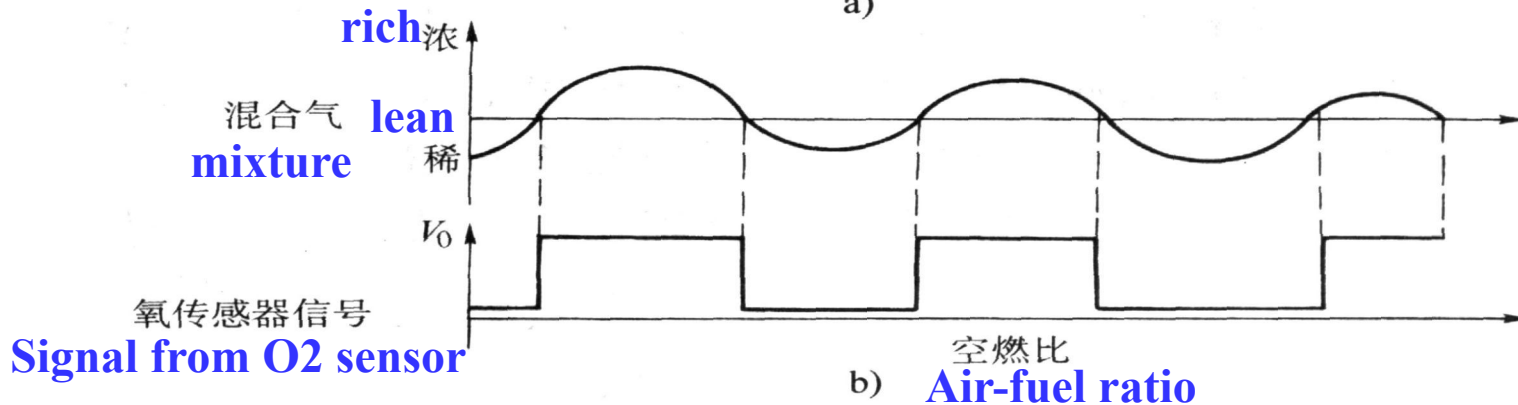


氧化锆具有在氧浓度差的作用下产生电动势的功能。大气与排气的氧浓度差所产生的电动势提供给ECU，作为改变燃油量的依据。

Zirconia has a function to generate an electromotive force because of the oxygen concentration difference.



a)

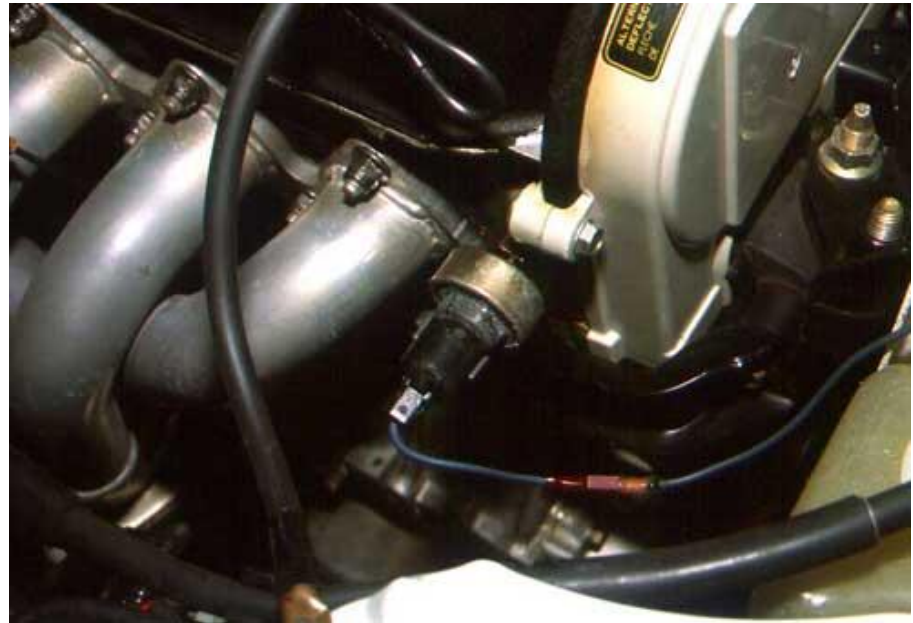


b)

Knock sensor 爆震传感器

爆震传感器作为点火定时控制的反馈元件用来检测发动机的爆燃强度，借以实现点火定时的闭环控制，以便有效地抑制发动机爆燃的发生。

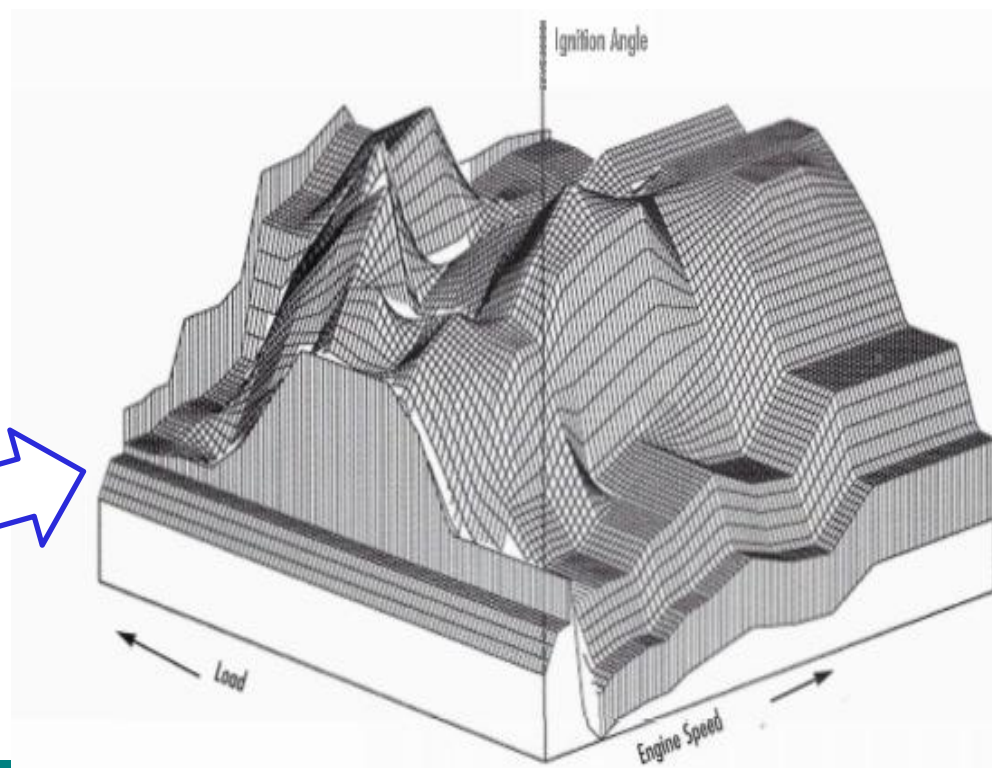
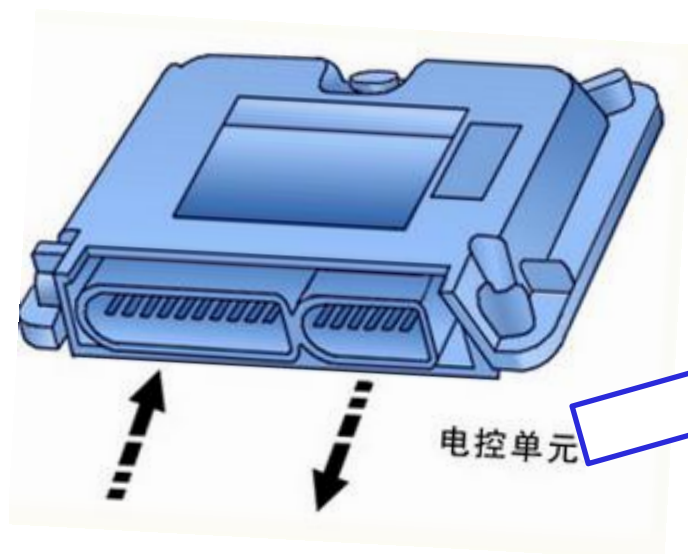
The knock sensor is used as a feedback element of the ignition timing control by detecting the strength of deflagrate of engine, in order to achieve the closed-loop control of ignition timing.

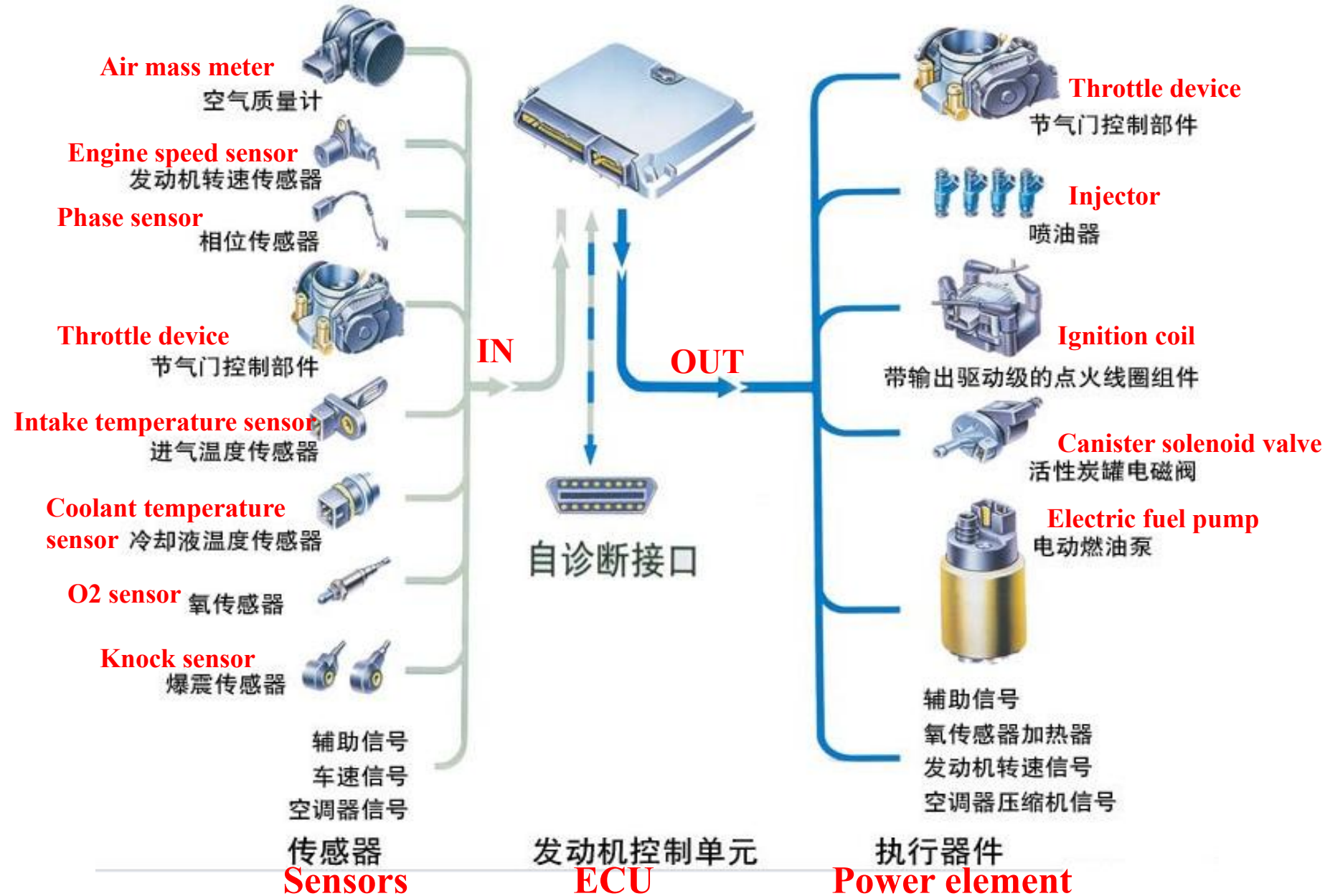


ECU 电控单元

ECU的功用是采集和处理各种传感器的输入信号，根据发动机工作的要求（喷油脉宽、点火提前角等），进行控制决策的运算，并输出相应的控制信号。

ECU comprehensively analyzes and calculates the signals from different kinds of sensors, and controls the mass of fuel injected, ignition timing and other controlling events.





Questions for Review

- 1. What means of 10%,50% and 90% distilled temperature for gasoline engine?**
- 2. What means for the number of gasoline ?**
- 3. What is knock in gasoline engine?**
- 4. How many parts consist of the fuel injection system for gasoline engine? And the difference of PFI and GDI?**
- 5. The main sensors and actuators for gasoline engine?**
- 6. The function of carbon canister?**
- 7. How to measure the air mass into the cylinder?**
- 8. How to measure the engine speed?**